Name

Mathematics Stage 5 Paths

Surds

Book 1

Version: 261008

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-IND-P-02 describes and performs operations with surds and fractional indices

Describe surds

- Describe a real number as a number that can be represented by a point on the number line
- Examine the differences between rational and irrational numbers and recognise that all rational and irrational numbers are real
- Convert between recurring decimals and their fractional form using digital tools
- Describe the term *surd* as referring to irrational expressions of the form $\sqrt[n]{x}$ where x is a rational number and n is an integer such that $n \geq 2$, and $x > 0$ when n is even
- Recognise that a surd is an exact value that can be approximated by a rounded decimal
- Demonstrate that $\sqrt{x}$ is undefined for $x < 0$ and that $\sqrt{x} = 0$ when $x = 0$ using digital tools
- Describe $\sqrt{x}$ as the positive square root of x for $x > 0$ and $\sqrt{0} = 0$

Apply knowledge of surds to solve problems

- Establish and apply the following results for $x > 0$ and $y > 0$: $\left(\sqrt{x}\right)^2 = x = \sqrt{x^2}$, $\sqrt{xy} = \sqrt{x} \times \sqrt{y}$ and $\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$
- Apply the 4 operations to simplify expressions involving surds
- Expand and simplify expressions involving surds
- Rationalise the denominators of surds of the form $\frac{a\sqrt{b}}{c\sqrt{d}}$

Recurring Decimals as Fractions

ⓘ Recurring Decimals as Fractions

Every terminating decimal can be written as a fraction using knowledge of place value:

$$0.827 = 827 \text{ thousandths}$$
$$= \frac{827}{1000}$$

Every recurring decimal can also be written as a fraction by looking at patterns.
You will learn a formal proof after learning algebraic equations.

$$0.\dot{3} \rightarrow \frac{1}{3}$$

FOUNDATION

1 $\frac{1}{3}$ as a decimal is 0.333333333333333…

a Express this recurring decimal using the correct notation.

$0.\dot{3}$

b Without using a calculator, what would $\frac{2}{3}$ be?

$0.\dot{6}$

c Check $2 \div 3$ on a calculator. Explain why it is different to your answer to part **b**.

The calculator ran out of digits in its display, so it rounded the last digit up to 7.

2 Convert to a decimal:

a $\frac{1}{3} = 0.333333333333333 \ldots = 0.\dot{3}$

b $\frac{2}{3} =$ $0.\dot{6}$

c $\frac{2}{6} =$ $0.\dot{3}$

d $\frac{3}{6} =$ 0.5

e $\frac{4}{6} =$ $0.\dot{6}$

f $\frac{6}{6} =$ 1

g $\frac{7}{6} =$ $1.1\dot{6}$

h $\frac{1}{7} =$ $0.\overline{142857}$

i $\frac{2}{7} =$ $0.\overline{285714}$

j $\frac{3}{7} =$ $0.\overline{428571}$

k $\frac{4}{7} =$ $0.\overline{571428}$

l $\frac{5}{7} =$ $0.\overline{714285}$

m $\frac{6}{7} =$ $0.\overline{857142}$

3 Using a calculator, convert these to decimals.

a $\frac{1}{9}$

$0.\dot{1}$

b $\frac{2}{9}$

$0.\dot{2}$

c $\frac{3}{9}$

$0.\dot{3}$

d $\frac{4}{9}$

$0.\dot{4}$

e $\frac{7}{9}$

$0.\dot{7}$

f $\frac{8}{9}$

$0.\dot{8}$

Using the pattern above, express these as fractions:

g $0.\dot{4}$

$\frac{4}{9}$

h $0.\dot{8}$

$\frac{8}{9}$

i $0.\dot{6}$

$\frac{6}{9} = \frac{2}{3}$

4 Using a calculator, convert these to decimals.

a $\frac{1}{99}$

$0.\overline{01}$

b $\frac{8}{99}$

$0.\overline{08}$

c $\frac{14}{99}$

$0.\overline{14}$

d $\frac{48}{99}$

$0.\overline{48}$

e $\frac{72}{99}$

$0.\overline{72}$

f $\frac{87}{99}$

$0.\overline{87}$

Using the pattern above, express these as fractions:

g $0.\overline{64}$

$\frac{64}{99}$

h $0.\overline{05}$

$\frac{5}{99}$

i $0.\overline{93}$

$\frac{93}{99} = \frac{31}{33}$

5 Using a calculator, convert these to decimals.

a $\frac{1}{999}$

$0.\overline{001}$

b $\frac{12}{999}$

$0.\overline{012}$

c $\frac{412}{999}$

$0.\overline{412}$

Using the pattern above, express these as fractions:

d $0.\overline{942}$

$\frac{942}{999} = \frac{314}{333}$

e $0.\overline{023}$

$\frac{23}{999}$

f $0.\overline{138}$

$\frac{138}{999}$

6 Convert these decimals to fractions.

a	$0.\dot{3}$	**g**	$0.1\dot{4}$	**m**	$1.0\overline{14}$
	$\frac{1}{3}$		$\frac{1}{10}+\frac{4}{90}=\frac{13}{90}$		$1\frac{144}{990}=1\frac{7}{445}$
b	$0.\overline{34}$	**h**	$0.01\dot{4}$	**n**	$2.\dot{4}$
	$\frac{34}{99}$		$\frac{1}{100}+\frac{4}{900}=\frac{13}{900}$		$2\frac{4}{9}$
c	$0.\overline{35}$	**i**	$0.\overline{014}$	**o**	$2.\overline{43}$
	$\frac{35}{99}$		$\frac{14}{999}$		$2\frac{43}{9}$
d	$0.\overline{357}$	**j**	$1.\dot{4}$	**p**	$3.\overline{43}$
	$\frac{357}{999}$		$1\frac{4}{9}$		$3\frac{43}{99}$
e	$0.\dot{4}$	**k**	$1.0\dot{4}$	**q**	$3.0\overline{43}$
	$\frac{4}{9}$		$1\frac{13}{90}$		$3\frac{43}{990}$
f	$0.0\dot{4}$	**l**	$1.\overline{14}$		
	$\frac{4}{90}=\frac{2}{45}$		$1\frac{14}{99}$		

7 Read the example carefully and answer the questions.

Example converting recurring decimals to fractions without using a calculator

Convert $0.\dot{6}$ to a fraction.

Let $x = 0.6666666\ldots$ (1)
$10x = 6.6666666\ldots$ (2)

Using subtraction: (2) – (1):

$$9x = 6$$
$$x = \frac{6}{9}$$
$$\therefore x = 0.6666666\ldots = \frac{6}{9}$$

Convert $0.0\dot{6}$ to a fraction.

Let $x = 0.06666666\ldots$ (1)
$100x = 6.6666666\ldots$ (2)

Using subtraction: (2) – (1):

$$99x = 6$$
$$x = \frac{6}{99}$$
$$\therefore x = 0.06666666\ldots = \frac{6}{99}$$

Convert these recurring decimals to fractions using the above method.

a $0.\dot{8}$	**b** $0.\dot{4}$	**c** $0.\dot{7}$
$x = 0.8888888\ldots$	$x = 0.4444444\ldots$	
$10x = 8.8888888\ldots$	$10x =$ ……	
$9x =$ ……	$9x =$ ……	
$\therefore x =$ ……	$\therefore x =$ ……	
$\frac{8}{9}$	$\frac{4}{9}$	$\frac{7}{9}$

d $1.\dot{2}$

$1\frac{2}{9}$

e $5.\dot{7}$

$5\frac{7}{9}$

f $8.\dot{2}$

$8\frac{2}{9}$

g $0.\overline{81}$

$x = 0.8181818181\ldots$

$100x = 81.8181818181\ldots$

$99x = \ldots\ldots\ldots\ldots\ldots$

$\therefore x = \ldots\ldots\ldots\ldots\ldots$

$\frac{9}{11}$

h $0.\overline{69}$

$\frac{23}{33}$

i $1.\overline{43}$

$1\frac{43}{99}$

j $3.\overline{72}$

$3\frac{8}{11}$

k $9.\overline{75}$

$9\frac{26}{33}$

l $0.\overline{132}$

$\frac{44}{333}$

m $9.\overline{04}$

$9\frac{4}{99}$

n $0.\overline{037}$

$\frac{37}{999}$

o $2.\overline{917}$

$2\frac{917}{999}$

p $6.\overline{0125}$

$6\frac{125}{9999}$

q $13.\overline{1375}$

$13\frac{1375}{9999}$

r $165.\overline{005}$

$165\frac{5}{999}$

8 By finding $10x$ and $100x$, show that 8.122222222222… = $\frac{731}{90}$

$x = 8.1222\ldots$

$10x = 81.222\ldots$

$100x = 812.222\ldots$

$90x = 731$

$x = \frac{731}{90}$

Convert these decimals to fractions.

a $0.3\dot{5}$

$\frac{16}{45}$

b $0.51\dot{2}$

$\frac{461}{900}$

c $0.0\dot{9}$

$\frac{1}{10}$

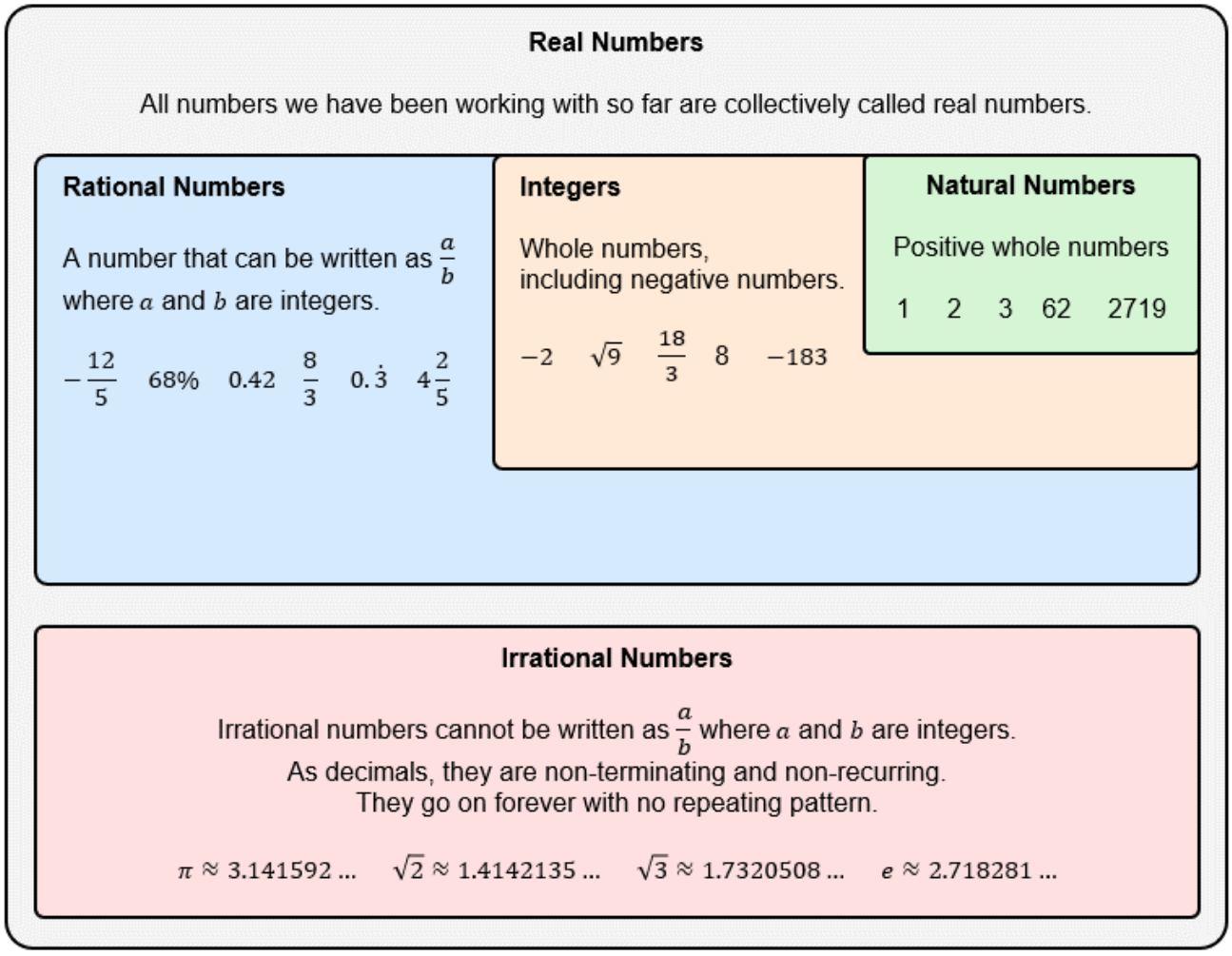
9 Convert these to fractions.

a	$0.\dot{1}$	$\frac{1}{9}$	**b**	$0.0\dot{1}$	$\frac{1}{90}$	**c**	$0.8\dot{1}$	$\frac{73}{90}$	**d**	$0.\overline{18}$	$\frac{18}{99} = \frac{2}{11}$	**e**	$0.1\overline{18}$	$\frac{13}{110}$
f	$0.\dot{2}$	$\frac{2}{9}$	**g**	$0.0\dot{2}$	$\frac{2}{90} = \frac{1}{45}$	**h**	$0.7\dot{2}$	$\frac{65}{90} = \frac{13}{18}$	**i**	$0.\overline{27}$	$\frac{27}{99} = \frac{3}{11}$	**j**	$0.5\overline{27}$	$\frac{29}{55}$
k	$0.\dot{3}$	$\frac{3}{9} = \frac{1}{3}$	**l**	$0.0\dot{3}$	$\frac{3}{90} = \frac{1}{30}$	**m**	$0.6\dot{3}$	$\frac{57}{90} = \frac{19}{30}$	**n**	$0.\overline{36}$	$\frac{36}{99} = \frac{4}{11}$	**o**	$0.4\overline{36}$	$\frac{24}{55}$
p	$0.\dot{4}$	$\frac{4}{9}$	**q**	$0.0\dot{4}$	$\frac{4}{90} = \frac{2}{45}$	**r**	$0.5\dot{4}$	$\frac{49}{90}$	**s**	$0.\overline{45}$	$\frac{45}{99} = \frac{5}{11}$	**t**	$0.3\overline{45}$	$\frac{19}{55}$
u	$0.\dot{5}$	$\frac{5}{9}$	**v**	$0.0\dot{5}$	$\frac{5}{90} = \frac{1}{18}$	**w**	$0.4\dot{5}$	$\frac{41}{90}$	**x**	$0.\overline{54}$	$\frac{54}{99} = \frac{6}{11}$	**y**	$0.9\overline{54}$	$\frac{21}{22}$

Irrational Numbers

ⓘ Rational vs Irrational

Rational numbers can be written as a fraction $\frac{a}{b}$ where a and b are integers and $b \neq 0$.	**Irrational** numbers cannot be written as a fraction, such as π and $\sqrt{2}$. As decimals, they are **non-terminating** and **non-recurring**.



Identify whether a number is rational or irrational

Using a calculator, represent these numbers as decimals and decide whether they are irrational or rational.

a	$\frac{1}{3}$	Rational	Irrational	**i**	$\sqrt{4}$	Rational	Irrational
b	3	Rational	Irrational	**j**	$\sqrt{5}$	Rational	Irrational
c	0.3	Rational	Irrational	**k**	$\sqrt{5}+1$	Rational	Irrational
d	$0.\dot{3}$	Rational	Irrational	**l**	$\frac{\sqrt{2}}{2}$	Rational	Irrational
e	$\sqrt{3}$	Rational	Irrational	**m**	$2\sqrt{2}$	Rational	Irrational
f	$\sqrt{9}$	Rational	Irrational	**n**	π	Rational	Irrational
g	0.25	Rational	Irrational	**o**	$-\frac{22.5}{7}$	Rational	Irrational
h	25%	Rational	Irrational	**p**	$3.\overline{14}$	Rational	Irrational

Key Ideas

1. Rational numbers can be written as a fraction $\frac{a}{b}$, where a and b are …… and $b \neq 0$
2. If a number can be written as a terminating or …… decimal, then it is rational.
3. Irrational numbers cannot be written as a fraction with integers, when written as a decimal, they are non-…… and non-……

Example Approximate irrational numbers

Approximate $\sqrt{2}$ to 2 decimal places.

$\sqrt{2} = 1.414213562\ldots$

≈ 1.41

Approximate e to 2 decimal places.

$e = 2.718281828\ldots$

≈ 2.72

Guided Practice

Approximate these numbers to 2 decimal places.

a $\sqrt{3}$

1.73

b π

3.14

c $\sqrt{25}$

5.00

d $\sqrt{18}$

4.24

e $\frac{1}{\sqrt{3}}$

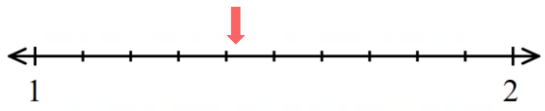
0.58

f $2\sqrt{2}$

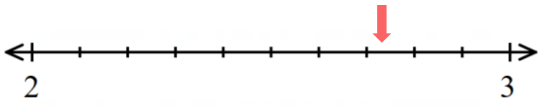
2.83

Example Locate the approximate location of irrational numbers

$\sqrt{2} \approx 1.41$



$e \approx 2.72$



Guided Practice

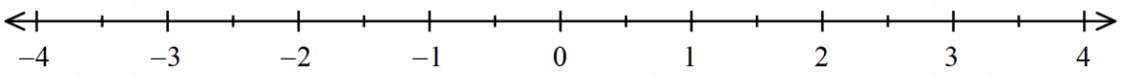
Using a calculator, locate the approximate location of these numbers on the number line.

a $\sqrt{3}$

b $-\sqrt{5}$

c $\frac{\sqrt{2}}{2}$

d $-\pi$



1 Circle which of these numbers are integers.

10 9.6 $\frac{3}{4}$ −3 117

10, −3, 117

Explain why all integers are rational numbers.

Integers can be written with a denominator of 1.

2 Circle which of these numbers are rational.

$\frac{7}{2}$ $\sqrt{3}$ 1.5 11

$\frac{7}{2}$, 1.5, 11

Explain how you know whether a number is rational.

The number can be written as a fraction with whole number numerator and denominator.

3 Explain why $\sqrt{4}$ is a rational number.

$\sqrt{4} = 2$, which is an integer, so it's rational.

4 Look at these numbers.

$\sqrt{2}$ $\sqrt{3}$ $\sqrt{4}$ $\sqrt{5}$ $3\sqrt{7}$ $2\sqrt{100}$ $-\sqrt{3}$ $-\sqrt{25}$

a Which numbers are irrational?

$\sqrt{3}, \sqrt{5}, 3\sqrt{7}, -\sqrt{3}$

b Which numbers are rational?

$\sqrt{4}, 2\sqrt{100}, -\sqrt{25}$

5 Emily says:



> 0.7777... recurring is irrational as you cannot write it as a fraction.

Explain why she is incorrect.

0.777… can be written as $\frac{7}{9}$, so it's rational.

6 Dolly says:



> π is a rational number, since it is roughly equal to $\frac{22}{7}$

Explain why Dolly is incorrect.

π is roughly equal to $\frac{22}{7}$ but not exactly. π is a non-terminating, non-recurring decimal, so it is irrational.

7 Becky says:



> $\sqrt{25}$ is not rational, because it is a square root.

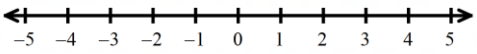
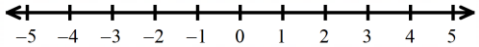
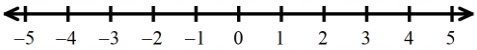
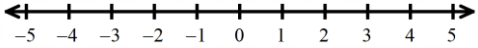
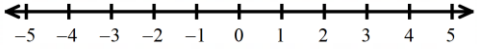
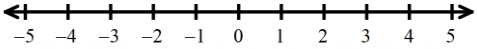
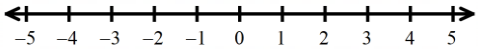
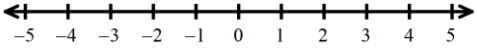
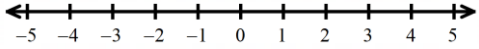
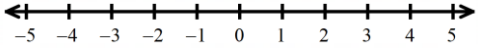
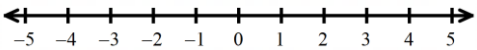
Explain why Becky is incorrect.

$\sqrt{25}$ simplifies to 5, which is rational.

8 Using a calculator, approximate these numbers to 2 decimal places.

a $\sqrt{5}$ 2.24	**b** $2\sqrt{5}$ 4.47	**c** 1.61803399 1.62
d $\frac{22}{7}$ 3.14	**e** $\frac{\pi}{2}$ 1.57	**f** e^2 7.39

9 Complete the table.

Number	Rational/ Irrational?	Approximation (2 d.p.)	Approximation (3 d.p.)	Approximate location on number line
e	Irrational	≈ 2.72	≈ 2.718	
$\sqrt{6}$	Irrational	2.45	2.449	
$2\sqrt{6}$	Irrational	4.90	4.899	
$\sqrt{24}$	Irrational	4.90	4.899	
$\sqrt{13}$	Irrational	3.61	3.606	
$\sqrt{16}$	Rational	4.00	4.000	
π	Irrational	3.14	3.142	
3π	Irrational	9.42	9.425	
$0.\dot{3}$	Rational	0.33	0.333	
$\frac{7}{3}$	Rational	2.33	2.333	
-5	Rational	-5.00	-5.000	
$-\sqrt{3}$	Irrational	-1.73	-1.732	

Surds

ⓘ Roots and Surds

- Square rooting $\sqrt{x}$ is the inverse operation of squaring x^2.
- We call a root a **surd** if it is irrational: E.g., $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt[3]{7}$
- It is best to leave surds in exact form, rather than a rounded decimal.
- It is impossible (for now) to square root a negative number $\sqrt{-4}$, we say it is **not real**.

🔍 **Identify** surds

Determine whether these roots are surds

a	$\sqrt{3}$	Surd	Rational	Not real
b	$2\sqrt{3}$	Surd	Rational	Not real
c	$\sqrt{4}$	Surd	Rational	Not real
d	$\sqrt{8}$	Surd	Rational	Not real
e	$\sqrt{16}$	Surd	Rational	Not real
f	$2\sqrt{16}$	Surd	Rational	Not real
g	$3+\sqrt{16}$	Surd	Rational	Not real
h	$\sqrt[3]{16}$	Surd	Rational	Not real
i	$\sqrt[3]{8}$	Surd	Rational	Not real
j	$\sqrt[3]{6}$	Surd	Rational	Not real
k	$\sqrt{6}$	Surd	Rational	Not real
l	$\sqrt{9}$	Surd	Rational	Not real
m	$-\sqrt{9}$	Surd	Rational	Not real
n	$\sqrt[3]{9}$	Surd	Rational	Not real
o	$\sqrt{0}$	Surd	Rational	Not real
p	$\sqrt{-0}$	Surd	Rational	Not real
q	$\sqrt{-1}$	Surd	Rational	Not real
r	$-\sqrt{1}$	Surd	Rational	Not real
s	$-\sqrt{-1}$	Surd	Rational	Not real
t	$\sqrt[3]{-8}$	Surd	Rational	Not real
u	$\sqrt{-(-25)}$	Surd	Rational	Not real

1 Tom says $\sqrt{1}$ is a surd

Katie says $\sqrt{2}$ is a surd

Who is correct?

A Only Tom **B** Only Katie **C** Both **D** Neither

2 Which of the following is **not** a surd?

A $\sqrt{45}$ **B** $\sqrt{47}$ **C** $\sqrt{49}$ **D** $\sqrt{51}$

3 Which of the following is **not** a surd?

A $\sqrt{84}$ **B** $\sqrt{83}$ **C** $\sqrt{82}$ **D** $\sqrt{81}$

4 Complete the sentences

a A number that cannot be expressed as a fraction with integer numerator and denominator is called ……

b A …… is an irrational root, for example $\sqrt{2}$ or $\sqrt[3]{5}$

c The decimal representation of a surd is a decimal that goes on …… and never ……

d $\sqrt{7}$ is a surd because its …… representation doesn't repeat and goes on forever.

e $\sqrt{25}$ is …… a surd because its decimal representation is ……

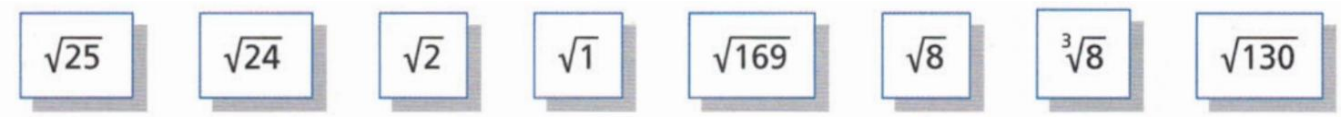
5 Jono says $\sqrt{18}$ is a surd. Explain why he is correct.

$\sqrt{18}$ cannot be expressed as a fraction with integers. It is a non-terminating, non-recurring decimal.

6 Becky says $\sqrt{25}$ is not a surd. Explain why she is correct.

$\sqrt{25}$ can be simplified to 5

7 Circle the surds.

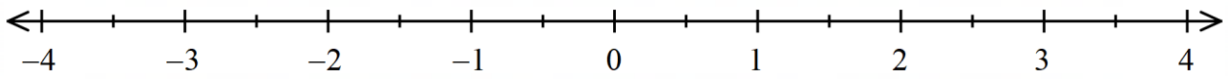


$\sqrt{24}, \sqrt{2}, \sqrt{8}, \sqrt{130}$

8 Express each number as a decimal using a calculator and then decide whether they are irrational or rational. Round the decimals to 2 d.p.

a $\sqrt{5}$ Irrational, 2.24	**b** 18% Rational, 0.18	**c** $\frac{2}{5}$ Rational, 0.4	**d** -124% Rational, -1.24
e $1\frac{5}{7}$ Rational, $1.\overline{714285}$	**f** $-\sqrt{2}$ Irrational, -1.41	**g** $2\sqrt{3}$ Irrational, 3.46	**h** π Irrational, 3.14

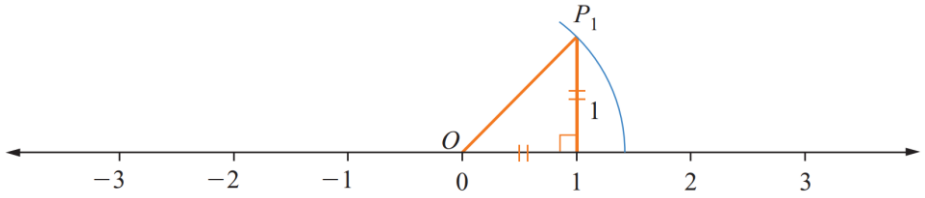
Show each number's approximate location on the number line.



9 Using a calculator, or otherwise, classify these numbers as either surds, rational, or not real.

a $\sqrt{7}$ surd	**b** $2\sqrt{11}$ surd	**c** $2\sqrt{25}$ rational	**d** $-5\sqrt{144}$ rational
e $\frac{3\sqrt{9}}{2}$ rational	**f** $\frac{-5\sqrt{3}}{2}$ surd	**g** $1-\sqrt{3}$ surd	**h** $2+\sqrt{4}$ rational
i $\sqrt{\frac{4}{25}}$ rational	**j** $\sqrt{\frac{5}{16}}$ surd	**k** $\sqrt{-11}$ Not real	**l** $\sqrt{99}$ surd

10 Consider the number line:



a Use Pythagoras' Theorem to calculate the length of OP_1

$\sqrt{2}$

b Using a pair of compasses, with point at O, the location of $\sqrt{2}$ on the number line has been accurately marked. What is the approximate value of $\sqrt{2}$?

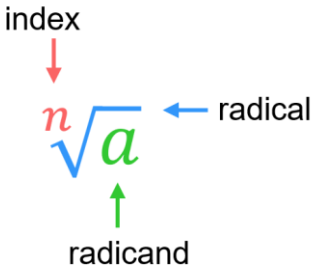
1.41

c $\sqrt{2}$ can be placed on a number line. Is it a real number?

Yes

Surd Laws

ⓘ Language of Surds



Identify parts of a surd

Surd	Index	Radicand
$\sqrt[3]{7}$		
$\sqrt{7}$		
$\sqrt[3]{a}$		
$3\sqrt{a}$		
$\sqrt[9]{n^2}$		
$\sqrt[9]{n+5}$		
$2\sqrt[6]{n^2+5}$		
	5	z^3

Surd Laws

Inverse Relationship	Surd Law of Multiplication	Surd Law of Division
$(\sqrt{x})^2 = x$ $\sqrt{x^2} = x$	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$	$\sqrt{a} \div \sqrt{b} = \sqrt{\frac{a}{b}}$

Multiply and Divide Surds using the Laws

$(\sqrt{11})^2$ $= 11$	$\sqrt{3} \times \sqrt{10}$ $= \sqrt{30}$	$\sqrt{24} \div \sqrt{8}$ $= \sqrt{3}$

Simplify these expressions.

a $\sqrt{3} \times \sqrt{7}$ $\sqrt{21}$	**b** $\sqrt{8} \times \sqrt{2}$ 4	**c** $\sqrt{2} \times \sqrt{11}$ $\sqrt{22}$	**d** $(\sqrt{12})^2$ 12
e $\sqrt{20} \div \sqrt{2}$ $\sqrt{10}$	**f** $\sqrt{45} \div \sqrt{5}$ 3	**g** $\sqrt{18} \div \sqrt{6}$ $\sqrt{3}$	**h** $\sqrt{49} \div \sqrt{7}$ $\sqrt{7}$

Identify whether you can use a surd law to simplify

a	$\sqrt{7} \times \sqrt{5}$	Yes | No	**e**	$\sqrt{6} \times \sqrt{0.5}$	Yes | No
b	$\sqrt{7} + \sqrt{5}$	Yes | No	**f**	$\sqrt{6} \div \sqrt{10}$	Yes | No
c	$\sqrt{7} \div \sqrt{5}$	Yes | No	**g**	$\sqrt{6} + \sqrt{12}$	Yes | No
d	$\sqrt{7} - \sqrt{7}$	Yes | No	**h**	$\sqrt{6} - \sqrt{3}$	Yes | No

Key Ideas

1. Raising to a power and rooting are ..... operations; they undo each other.
2. $\sqrt{x^2} = (\sqrt{x})^2 =$ .....
3. The surd law of multiplication says that $\sqrt{a} \times \sqrt{b} =$ .....
4. The surd law of division says that $\frac{\sqrt{a}}{\sqrt{b}} =$ .....

1 Evaluate the following.

a $(\sqrt{11})^2$ — 11

b $\sqrt{11^2}$ — 11

c $(\sqrt{8})^2$ — 8

d $\sqrt{8^2}$ — 8

e $(\sqrt{6})^2$ — 6

2

a Write down the value of

i $\sqrt{4}$ — 2

ii $\sqrt{25}$ — 5

iii $\sqrt{100}$ — 10

ii $\sqrt{400}$ — 20

b Hence show that

i $\sqrt{4} \times \sqrt{100} = \sqrt{400}$ — $2 \times 10 = 20$

ii $\sqrt{100} \div \sqrt{4} = \sqrt{25}$ — $10 \div 2 = 5$

3

a Write down the value of

i $\sqrt{9}$ — 3

ii $\sqrt{16}$ — 4

b Hence show that

i $\sqrt{9} + \sqrt{16} \neq \sqrt{25}$ — $3 + 4 \neq 5$

ii $\sqrt{16} - \sqrt{9} \neq \sqrt{7}$ — $4 - 3 \neq \sqrt{7}$

4 Use the fact that $\sqrt{a} \times \sqrt{b} = \sqrt{ab}$ to write each expression as a single surd.

a $\sqrt{5} \times \sqrt{3}$ — $\sqrt{15}$	**b** $\sqrt{7} \times \sqrt{11}$ — $\sqrt{77}$	**c** $\sqrt{51} \times \sqrt{2}$ — $\sqrt{102}$	**d** $\sqrt{15} \times \sqrt{21}$ — $\sqrt{315}$
e $\sqrt{2} \times \sqrt{11}$ — $\sqrt{22}$	**f** $\sqrt{3} \times \sqrt{7}$ — $\sqrt{21}$	**g** $\sqrt{10} \times \sqrt{3}$ — $\sqrt{30}$	**h** $\sqrt{10} \times \sqrt{7}$ — $\sqrt{70}$

5 Use the fact that $\sqrt{a} \div \sqrt{b} = \sqrt{\frac{a}{b}}$ to write each expression as a single surd.

a $\sqrt{21} \div \sqrt{3}$ — $\sqrt{7}$	**b** $\sqrt{26} \div \sqrt{2}$ — $\sqrt{13}$	**c** $\sqrt{51} \div \sqrt{17}$ — $\sqrt{3}$	**d** $\sqrt{3} \div \sqrt{21}$ — $\sqrt{\frac{1}{7}}$
e $\frac{\sqrt{20}}{\sqrt{2}}$ — $\sqrt{10}$	**f** $\frac{\sqrt{45}}{\sqrt{5}}$ — 3	**g** $\frac{\sqrt{32}}{\sqrt{8}}$ — 2	**h** $\frac{\sqrt{35}}{\sqrt{5}}$ — $\sqrt{7}$

6

a Using your knowledge of index laws, explain why $(3\sqrt{5})^2 = 3^2(\sqrt{5})^2$

Index law of power of a product: $(ab)^n = a^n b^n$

b Therefore, explain why $(3\sqrt{5})^2 = 45$

$3^2\sqrt{5}^2 = 9(5) = 45$

7 Evaluate the following.

a $(3\sqrt{2})^2$ 18

b $(2\sqrt{3})^2$ 12

c $(5\sqrt{2})^2$ 50

d $(10\sqrt{7})^2$ 700

e $(4\sqrt{5})^2$ 80

8 Simplify these expressions. Write your answer as a whole number.

a $(\sqrt{2})^2$ 2

b $(\sqrt{3})^2$ 3

c $(\sqrt{5})^2$ 5

d $(\sqrt{10})^2$ 10

e $(2\sqrt{10})^2$ 40

f $(3\sqrt{10})^2$ 90

g $3(\sqrt{10})^2$ 30

h $5(\sqrt{10})^2$ 50

i $10(\sqrt{10})^2$ 100

Simplifying Surds

ⓘ Simplify surds

1. Factorise the **radicand** (number inside the root) according to its **highest square factor**.

   $4,\quad 9,\quad 16,\quad 25,\quad 36,\quad 49,\quad 64,\quad 81,\quad 100,\quad 121,\quad 144\ldots$

2. Split the surd, then simplify perfect squares.

Example Simplify surds

$\sqrt{45}$

1. $\sqrt{9\times 5}$
2. $\sqrt{9}\times\sqrt{5}$

$=3\sqrt{5}$

$3\sqrt{40}$

1. $3\sqrt{4\times 10}$
2. $3\sqrt{4}\sqrt{10}$

$=6\sqrt{10}$

$\frac{\sqrt{80}}{3}$

1. $\frac{\sqrt{16\times 5}}{3}$
2. $\frac{4\sqrt{5}}{3}$

Guided Practice

Simplify these surds.

a $\sqrt{12}$ $2\sqrt{3}$	**b** $\sqrt{28}$ $2\sqrt{7}$	**c** $\sqrt{54}$ $3\sqrt{6}$	**d** $3\sqrt{8}$ $6\sqrt{2}$
e $\frac{\sqrt{48}}{2}$ $2\sqrt{3}$	**f** $\frac{5\sqrt{40}}{6}$ $\frac{5\sqrt{10}}{3}$	**g** $\sqrt{\frac{75}{9}}$ $\frac{5}{\sqrt{3}}$ or $\frac{5\sqrt{3}}{3}$	**h** $\sqrt{\frac{50}{16}}$ $\frac{5}{2\sqrt{2}}$ or $\frac{5\sqrt{2}}{4}$

1 Simplify these surds

a $\sqrt{12}$ $= \sqrt{4 \times 3}$ $=$ $2\sqrt{3}$	**b** $\sqrt{45}$ $3\sqrt{5}$	**c** $\sqrt{24}$ $2\sqrt{6}$	**d** $\sqrt{48}$ $4\sqrt{3}$
e $\sqrt{75}$ $5\sqrt{3}$	**f** $\sqrt{500}$ $10\sqrt{5}$	**g** $\sqrt{98}$ $7\sqrt{2}$	**h** $\sqrt{90}$ $3\sqrt{10}$
i $\sqrt{128}$ $8\sqrt{2}$	**j** $\sqrt{360}$ $6\sqrt{10}$	**k** $\sqrt{162}$ $9\sqrt{2}$	**l** $\sqrt{80}$ $4\sqrt{5}$

2 There are many ways to factorise $\sqrt{600}$. 1 is given below. Find 2 more different ways.

$\sqrt{600} = 2\sqrt{150}$

$5\sqrt{24}, 10\sqrt{6}$

What is the simplest form of $\sqrt{600}$?

$10\sqrt{6}$

3 Work out each value of x

s $\sqrt{150} = 5\sqrt{x}$ 6	**t** $\sqrt{300} = x\sqrt{3}$ 10	**u** $\sqrt{x} = 7\sqrt{13}$ 637

4 Samira has simplified $\sqrt{200}$: $\sqrt{200} = \sqrt{4} \times \sqrt{50} = 2 \times \sqrt{50} = 2\sqrt{50}$

Explain how Samira can improve her answer.

Further simplify $\sqrt{50}$; $2\sqrt{50} = 2 \times 5\sqrt{2} = 10\sqrt{2}$

5 Consider the working out to simplify $\sqrt{72}$

$$\begin{aligned}\sqrt{72} &= \sqrt{9 \times 8} \\ &= 3\sqrt{8} \\ &= 3\sqrt{4 \times 2} \\ &= 3 \times 2 \times \sqrt{2} \\ &= 6\sqrt{2}\end{aligned}$$

a Explain why the working out for simplifying $\sqrt{72}$ is longer than it needs to be.

You can split $\sqrt{72}$ to $\sqrt{36}\sqrt{2}$;

9 was not the largest square factor of 72.

b Therefore, simplify $\sqrt{72}$ using as few steps as possible.

$\sqrt{72} = \sqrt{36}\sqrt{2} = 6\sqrt{2}$

6 Simplify these surds. Try to write as little working out as possible.

a $\sqrt{8}$ $2\sqrt{2}$

b $\sqrt{12}$ $2\sqrt{3}$

c $\sqrt{32}$ $4\sqrt{2}$

d $\sqrt{64}$ 8

e $\sqrt{128}$ $8\sqrt{2}$

f $\sqrt{192}$ $8\sqrt{3}$

g $\sqrt{320}$ $8\sqrt{5}$

h $\sqrt{80}$ $4\sqrt{5}$

i $\sqrt{40}$ $2\sqrt{10}$

j $\sqrt{90}$ $3\sqrt{10}$

k $\sqrt{160}$ $4\sqrt{10}$

l $\sqrt{1600}$ 40

m $\sqrt{200}$ $10\sqrt{2}$

n $\sqrt{250}$ $5\sqrt{10}$

o $\sqrt{175}$ $5\sqrt{7}$

7 Simplify these surds.

a $2\sqrt{18}$

$= 2 \times 3\sqrt{2}$

$= 6\sqrt{2}$

b $4\sqrt{48}$

$= 4 \times 4\sqrt{3}$

$16\sqrt{3}$

c $3\sqrt{98}$ $21\sqrt{2}$

d $4\sqrt{125}$ $20\sqrt{5}$

8 Simplify these surds.

a $\sqrt{8}$

$2\sqrt{2}$

b $2\sqrt{8}$

$4\sqrt{2}$

c $3\sqrt{8}$

$6\sqrt{2}$

d $\sqrt{72}$

$6\sqrt{2}$

f $\sqrt{72}$

$6\sqrt{2}$

g $5\sqrt{72}$

$30\sqrt{2}$

h $5\sqrt{144}$

60

i $5\sqrt{1440}$

$60\sqrt{10}$

k $5\sqrt{720}$

$60\sqrt{5}$

l $5\sqrt{360}$

$30\sqrt{10}$

m $5\sqrt{36}$

30

n $5\sqrt{25}$

25

9 Determine the exact side length, in simplest form, of a square with the given area.

a 32 m^2

$4\sqrt{2}$ m

b 120 cm^2

$2\sqrt{30}$ cm

c 240 mm^2

$4\sqrt{15}$ mm

10 Determine the exact radius, in simplest form, of a circle with the given area. $A = \pi r^2$

a 24π m^2

$2\sqrt{6}$ m

b 54π cm^2

$3\sqrt{6}$ cm

c 128π mm^2

$8\sqrt{2}$ mm

11 Simplify.

a $\sqrt{675}$

$15\sqrt{3}$

b $\sqrt{1183}$

$13\sqrt{7}$

c $\sqrt{2883}$

$31\sqrt{3}$

Sorry :)

12 Express these surds as a square root of a positive integer.

a $2\sqrt{3}$

$= \sqrt{4} \times \sqrt{3}$

$= \sqrt{12}$

b $4\sqrt{2}$

$= \sqrt{16} \times \sqrt{2}$

$= \ldots\ldots\ldots\ldots$

$\sqrt{32}$

c $3\sqrt{3}$

$\sqrt{27}$

d $3\sqrt{5}$

$\sqrt{45}$

e $6\sqrt{3}$

$\sqrt{108}$

f $8\sqrt{2}$

$\sqrt{128}$

g $10\sqrt{7}$

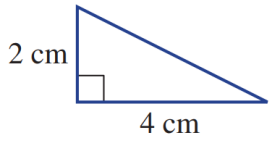
$\sqrt{700}$

h $9\sqrt{10}$

$\sqrt{810}$

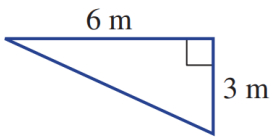
13 Use Pythagoras' Theorem to find the unknown length in simplest form.

a



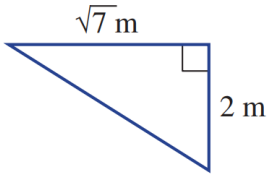
$2\sqrt{5}$ cm

b



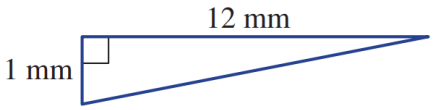
$3\sqrt{5}$ m

c



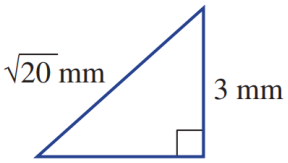
$\sqrt{11}$ m

d



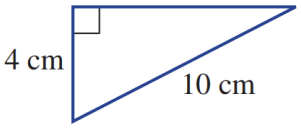
$\sqrt{145}$ mm

e



$\sqrt{11}$ mm

f



$2\sqrt{21}$ cm

14 Simplify the following.

a $\sqrt[3]{125} - \sqrt[3]{8}$

3

b $\sqrt[3]{24}$

$2\sqrt[3]{3}$

c $\sqrt[3]{2000}$

$10\sqrt[3]{2}$

d $\sqrt[5]{64}$

$2\sqrt[5]{2}$

15 Simplify the following.

a $\sqrt{a^2b}$

$a\sqrt{b}$

b $\sqrt{a^4b^2}$

a^2b

c $\sqrt[3]{a^6b^3}$

a^2b

d $\sqrt{a^5b^3}$

$a^2b\sqrt{ab}$

Addition and Subtraction of Surds

Addition and Subtraction of Surds

We can only add and subtract "**like**" surds; like surds have the same radicand and index.

1. Simplify all surds.
2. Collect like surds.

Example Add and subtract surds

$3\sqrt{7} - \sqrt{6} + 2\sqrt{7}$

2. $3\sqrt{7} - \sqrt{6} + 2\sqrt{7}$

$= 5\sqrt{7} - \sqrt{6}$

$\sqrt{44} + \sqrt{99}$

1. $\sqrt{4}\sqrt{11} + \sqrt{9}\sqrt{11}$

$2\sqrt{11} + 3\sqrt{11}$

2. $5\sqrt{11}$

Guided Practice

Simplify these expressions.

a	$2\sqrt{11} + 6\sqrt{11}$	**b**	$3\sqrt{6} + \sqrt{6}$	**c**	$5\sqrt{7} + 3\sqrt{5} - 4\sqrt{7}$
	$8\sqrt{11}$		$4\sqrt{6}$		$\sqrt{7} + 3\sqrt{5}$
d	$\sqrt{8} + \sqrt{2}$	**e**	$\sqrt{12} - \sqrt{3}$	**f**	$\sqrt{54} + \sqrt{24}$
	$3\sqrt{2}$		$\sqrt{3}$		$5\sqrt{6}$
g	$2\sqrt{8} - \sqrt{18}$	**h**	$2\sqrt{5} + \sqrt{45}$	**i**	$\sqrt{72} - \sqrt{50} + \sqrt{12}$
	$\sqrt{2}$		$5\sqrt{5}$		$2\sqrt{3} + \sqrt{2}$

Key Ideas

1. We can only add or subtract …… surds; surds with the same …… and ……
2. Sometimes we can create like surds by …… them.

1 Decide whether these pairs of numbers are like or unlike surds.

a	$3\sqrt{4},\quad 4\sqrt{2}$	**b**	$5\sqrt{3},\quad 2\sqrt{3}$	**c**	$4\sqrt{2},\quad 4\sqrt{7}$	**d**	$\sqrt{3},\quad 3\sqrt{5}$
	unlike		like		unlike		unlike
e	$6\sqrt{6},\quad 3\sqrt{3}$	**f**	$\sqrt{8},\quad -\sqrt{8}$	**g**	$19\sqrt{2},\quad -2\sqrt{2}$	**h**	$-3\sqrt{6},\quad 3\sqrt{5}$
	unlike		like		like		unlike

2 Complete the following:

a $11x - 5x =$ ....., so $11\sqrt{2} - 5\sqrt{2} =$ .....

$6x,\quad 6\sqrt{2}$

b $8x - x =$ ....., so $8\sqrt{2} - \sqrt{2} =$ .....

$7x,\quad 7\sqrt{2}$

c $8x - 7x =$ ....., so $8\sqrt{2} - 7\sqrt{2} =$ .....

$x,\quad \sqrt{2}$

3 Simplify these expressions.

a	$2\sqrt{5} + 4\sqrt{5}$	**b**	$5\sqrt{3} - 2\sqrt{3}$	**c**	$7\sqrt{2} - 3\sqrt{2}$
	$6\sqrt{5}$		$3\sqrt{3}$		$4\sqrt{2}$
d	$8\sqrt{2} - 5\sqrt{2}$	**e**	$4\sqrt{10} + 3\sqrt{10} - \sqrt{10}$	**f**	$\sqrt{21} - 5\sqrt{21} + 2\sqrt{21}$
	$3\sqrt{2}$		$6\sqrt{10}$		$-2\sqrt{21}$

4 Simplify the surd $\sqrt{48}$ .....

$4\sqrt{3}$

Hence, simplify the following:

a	$\sqrt{3} + \sqrt{48}$	**b**	$\sqrt{48} - 7\sqrt{3}$	**c**	$5\sqrt{48} - 3\sqrt{3}$
	$5\sqrt{3}$		$-3\sqrt{3}$		$17\sqrt{3}$

5 Simplify all surds, then collect like surds.

a	$\sqrt{8} - \sqrt{2}$ $= 2\sqrt{2} - \sqrt{2}$ $=$	**b**	$\sqrt{8} + 3\sqrt{2}$	**c**	$\sqrt{27} + \sqrt{3}$
	$\sqrt{2}$		$5\sqrt{2}$		$4\sqrt{3}$
d	$\sqrt{20} - \sqrt{5}$	**e**	$4\sqrt{18} - 5\sqrt{2}$	**f**	$2\sqrt{75} + 2\sqrt{3}$
	$\sqrt{5}$		$7\sqrt{2}$		$12\sqrt{3}$
g	$3\sqrt{44} + 2\sqrt{11}$	**h**	$3\sqrt{8} - \sqrt{18}$	**i**	$2\sqrt{125} - 3\sqrt{45}$
	$8\sqrt{11}$		$3\sqrt{2}$		$\sqrt{5}$

6 Simplify these expressions.

a $2\sqrt{3} + 3\sqrt{2} - \sqrt{3} + 2\sqrt{3}$

$3\sqrt{3} + 3\sqrt{2}$

b $3\sqrt{5} - 4\sqrt{2} + \sqrt{5} - 3\sqrt{2}$

$4\sqrt{5} - 7\sqrt{2}$

c $5\sqrt{2} + 2\sqrt{5} - 7\sqrt{2} - \sqrt{5}$

$-2\sqrt{2} + \sqrt{5}$

d $2\sqrt{3} + 2\sqrt{7} + 2\sqrt{3} - 2\sqrt{7}$

$4\sqrt{3}$

e $5\sqrt{11} + 3\sqrt{6} - 3\sqrt{11}$

$2\sqrt{11} + 3\sqrt{6}$

f $-4\sqrt{10} - 5\sqrt{2} + \sqrt{10}$

$-3\sqrt{10} - 5\sqrt{2}$

7 Simplify these expressions.

a $2\sqrt{3} + 5\sqrt{3}$

$7\sqrt{3}$

b $2\sqrt{3} + 5\sqrt{6}$

Can't simplify

c $2\sqrt{3} + 5\sqrt{12}$

$12\sqrt{3}$

d $2\sqrt{3} + 5\sqrt{27}$

$17\sqrt{3}$

e $2\sqrt{3} + 6\sqrt{27}$

$20\sqrt{3}$

f $2\sqrt{30} + 6\sqrt{27}$

$2\sqrt{30} + 18\sqrt{3}$

g $2\sqrt{30} + 6\sqrt{120}$

$14\sqrt{30}$

h $6\sqrt{30} + 2\sqrt{120}$

$10\sqrt{30}$

i $6\sqrt{60} + 2\sqrt{240}$

$20\sqrt{15}$

j $6\sqrt{6} + 2\sqrt{24}$

$10\sqrt{6}$

k $6\sqrt{6} + 2\sqrt{24} + 5\sqrt{36}$

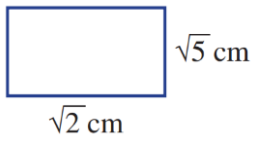
$10\sqrt{6} + 30$

l $6\sqrt{16} + 2\sqrt{24} + 5\sqrt{36}$

$54 + 4\sqrt{6}$

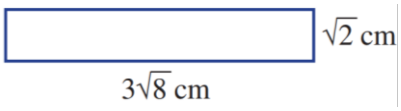
8 Calculate the perimeter in simplest form.

a



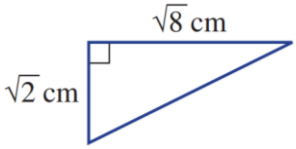
$2\sqrt{2} + 2\sqrt{5}$

b



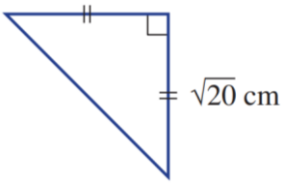
$14\sqrt{2}$

c



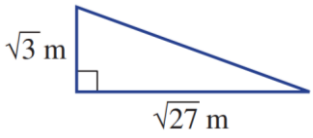
$3\sqrt{2} + \sqrt{10}$

d



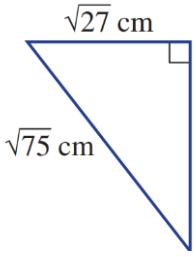
$4\sqrt{5} + 2\sqrt{10}$

e



$4\sqrt{3} + \sqrt{30}$

f



$12\sqrt{3}$

9 Prove that each of the following simplifies to zero by showing all steps.

h $5\sqrt{3} - \sqrt{108} + \sqrt{3}$

i $6\sqrt{2} - 2\sqrt{32} + 2\sqrt{2}$

j $2\sqrt{20} - 7\sqrt{5} + \sqrt{45}$

k $3\sqrt{2} - 2\sqrt{27} - \sqrt{50} + 6\sqrt{3} + \sqrt{8}$

10 Simplify the surds and combine the fractions using a common denominator.

a $\frac{\sqrt{8}}{3} - \frac{\sqrt{2}}{5}$ $\frac{7\sqrt{2}}{15}$

b $\frac{\sqrt{12}}{4} + \frac{\sqrt{3}}{6}$ $\frac{2\sqrt{3}}{3}$

c $\frac{3\sqrt{5}}{4} - \frac{\sqrt{20}}{3}$ $\frac{\sqrt{5}}{12}$

d $\frac{\sqrt{98}}{4} - \frac{5\sqrt{2}}{2}$ $-\frac{3\sqrt{2}}{4}$

e $\frac{2\sqrt{75}}{5} - \frac{3\sqrt{3}}{2}$ $\frac{\sqrt{3}}{2}$

f $\frac{\sqrt{63}}{9} - \frac{4\sqrt{7}}{5}$ $\frac{-21\sqrt{7}}{45}$

g $\frac{2\sqrt{18}}{3} - \frac{\sqrt{72}}{2}$ $-\sqrt{2}$

h $\frac{\sqrt{54}}{4} + \frac{\sqrt{24}}{7}$ $\frac{29\sqrt{6}}{28}$

i $\frac{\sqrt{27}}{5} - \frac{\sqrt{108}}{10}$ 0

j $\frac{5\sqrt{48}}{6} + \frac{2\sqrt{147}}{3}$ $8\sqrt{3}$

k $\frac{2\sqrt{96}}{5} - \frac{\sqrt{600}}{7}$ $\frac{34\sqrt{6}}{35}$

l $\frac{3\sqrt{125}}{14} - \frac{2\sqrt{80}}{21}$ $\frac{29\sqrt{5}}{42}$

16 Simplify fully.

a $\sqrt{600} + \sqrt{300} - \sqrt{216}$

$4\sqrt{6} + 10\sqrt{3}$

b $4\sqrt{18} + 3\sqrt{12} - 2\sqrt{50}$

$2\sqrt{2} + 6\sqrt{3}$

c $2\sqrt{175} - 5\sqrt{140} - 3\sqrt{28}$

$4\sqrt{7} - 10\sqrt{35}$

17 Find the value of x.

a $\sqrt{63} - \sqrt{28} = \sqrt{x}$

7

b $\sqrt{80} - \sqrt{20} = \sqrt{x}$

20

c $2\sqrt{150} - 3\sqrt{24} = \sqrt{x}$

96

Multiplying Surds

Multiplying Surds

$$a\sqrt{x} \times b\sqrt{y} = ab\sqrt{xy}$$

1. Multiply the coefficients and the radicands separately.
2. Simplify, if possible.

Example Multiply surds

$\sqrt{15} \times \sqrt{5}$	$5\sqrt{6} \times 7\sqrt{10}$
1. $\sqrt{75}$	1. $35\sqrt{60}$
2. $5\sqrt{3}$	2. $70\sqrt{15}$

Guided Practice

Simplify these expressions.

a $\sqrt{7} \times \sqrt{3}$ $\sqrt{21}$	**b** $\sqrt{5} \times \sqrt{20}$ 10	**c** $\sqrt{12} \times \sqrt{8}$ $4\sqrt{6}$
d $2 \times 3\sqrt{2}$ $6\sqrt{2}$	**e** $2\sqrt{6} \times 5\sqrt{3}$ $30\sqrt{2}$	**f** $-4\sqrt{5} \times 3\sqrt{11}$ $-12\sqrt{55}$
g $2\sqrt{3} \times 3\sqrt{15}$ $18\sqrt{5}$	**h** $(2\sqrt{5})^2$ 20	**i** $3\sqrt{2} \times -4\sqrt{6}$ $-24\sqrt{3}$

Key Ideas

1. $x\sqrt{m} \times y\sqrt{n} =$ .....
2. Remember to ..... the result as much as possible.

1 Simplify these expressions.

a $\sqrt{3} \times \sqrt{5}$ $\sqrt{15}$

b $\sqrt{7} \times \sqrt{3}$ $\sqrt{21}$

c $\sqrt{2} \times \sqrt{13}$ $\sqrt{26}$

d $\sqrt{2} \times (-\sqrt{15})$ $-\sqrt{30}$

e $-\sqrt{6} \times \sqrt{5}$ $-\sqrt{30}$

f $-\sqrt{6} \times (-\sqrt{11})$ $\sqrt{66}$

g $\sqrt{20} \div \sqrt{2}$ $\sqrt{10}$

h $\sqrt{18} \div \sqrt{3}$ $\sqrt{6}$

i $\sqrt{30} \div (-\sqrt{6})$ $-\sqrt{5}$

j $\frac{\sqrt{15}}{\sqrt{5}}$ $\sqrt{3}$

k $\frac{\sqrt{30}}{\sqrt{3}}$ $\sqrt{10}$

l $-\frac{\sqrt{26}}{\sqrt{2}}$ $-\sqrt{13}$

2 Simplify the following. Ensure your final answer is fully simplified.

a $\sqrt{9} \times \sqrt{9}$ 9

b $\sqrt{14} \times \sqrt{7}$ $7\sqrt{2}$

c $\sqrt{2} \times \sqrt{22}$ $2\sqrt{11}$

d $\sqrt{3} \times \sqrt{18}$ $3\sqrt{6}$

e $\sqrt{10} \times \sqrt{5}$ $5\sqrt{2}$

f $\sqrt{12} \times \sqrt{8}$ $4\sqrt{6}$

g $(2\sqrt{3})^2$ 12

h $(7\sqrt{3})^2$ 147

i $(9\sqrt{2})^2$ 162

3 Simplify $\sqrt{8} \times \sqrt{27}$ by:

a Simplifying each surd then multiplying

$2\sqrt{2} \times 3\sqrt{3}$
$= 6\sqrt{6}$

b Multiplying the surds then simplifying.

$\sqrt{216}$
$= \sqrt{36}\sqrt{6}$
$= 6\sqrt{6}$

Why is simplifying before multiplying easier in the example above?

You don't need to find square factors of a large number.

Simplify these expressions by simplifying each surd before multiplying.

c $\sqrt{8} \times \sqrt{18}$
$= 2\sqrt{\ } \times 3\sqrt{\ }$
$=$ 12

d $\sqrt{54} \times \sqrt{8}$ $12\sqrt{3}$

e $\sqrt{75} \times \sqrt{3}$ 15

f $\sqrt{12} \times \sqrt{18}$ $6\sqrt{6}$

4 Simplify these expressions.

a	$\sqrt{2} \times \sqrt{3}$	**f**	$\sqrt{12} \times \sqrt{3}$	**k**	$\sqrt{20} \times \sqrt{4}$
	$\sqrt{6}$		6		$4\sqrt{5}$
b	$\sqrt{4} \times \sqrt{3}$	**g**	$\sqrt{12} \times \sqrt{4}$	**l**	$\sqrt{20} \times \sqrt{5}$
	$2\sqrt{3}$		$4\sqrt{3}$		10
c	$\sqrt{5} \times \sqrt{3}$	**h**	$\sqrt{11} \times \sqrt{4}$	**m**	$\sqrt{20} \times \sqrt{10}$
	$\sqrt{15}$		$2\sqrt{11}$		$10\sqrt{2}$
d	$\sqrt{6} \times \sqrt{3}$	**i**	$\sqrt{10} \times \sqrt{4}$	**n**	$\sqrt{20} \times \sqrt{20}$
	$3\sqrt{2}$		$2\sqrt{10}$		20
e	$\sqrt{24} \times \sqrt{3}$	**j**	$\sqrt{10} \times \sqrt{8}$	**o**	$\sqrt{20} \times \sqrt{40}$
	$6\sqrt{2}$		$4\sqrt{5}$		$20\sqrt{2}$

5 Simplify these expressions.

a	$2\sqrt{2} \times \sqrt{3}$	**f**	$2\sqrt{12} \times 3\sqrt{3}$	**k**	$\sqrt{20} \times 2\sqrt{4}$
	$2\sqrt{6}$		36		$8\sqrt{5}$
b	$2\sqrt{4} \times \sqrt{3}$	**g**	$2\sqrt{12} \times 3\sqrt{4}$	**l**	$5\sqrt{20} \times 3\sqrt{5}$
	$4\sqrt{3}$		$24\sqrt{3}$		150
c	$\sqrt{5} \times 2\sqrt{3}$	**h**	$2\sqrt{11} \times 3\sqrt{4}$	**m**	$5\sqrt{20} \times 3\sqrt{10}$
	$2\sqrt{15}$		$12\sqrt{11}$		$150\sqrt{2}$
d	$\sqrt{6} \times 2\sqrt{3}$	**i**	$\sqrt{10} \times 2\sqrt{4}$	**n**	$5\sqrt{20} \times 3\sqrt{20}$
	$6\sqrt{2}$		$4\sqrt{10}$		300
e	$\sqrt{24} \times 3\sqrt{3}$	**j**	$\sqrt{10} \times 2\sqrt{8}$	**o**	$2\sqrt{20} \times 3\sqrt{40}$
	$18\sqrt{2}$		$8\sqrt{5}$		$120\sqrt{2}$

6 Consider the method of **factorising** a surd before multiplying:

Example Factorise a surd before multiplying

$\sqrt{7} \times \sqrt{14}$ $= \sqrt{7} \times \sqrt{2}\sqrt{7}$ $= 7\sqrt{2}$	The "normal" method would have been to multiply 7 and 14, then find the largest square factor. Why is this method faster? This method avoids working with large numbers.

Use the method of factorising to simply these surd multiplications.

a $\sqrt{8} \times \sqrt{24}$

$= \sqrt{8} \times \sqrt{\ }\sqrt{8}$

$=$

$8\sqrt{3}$

b $\sqrt{12} \times \sqrt{18}$

$= \sqrt{\ }\sqrt{6} \times \sqrt{\ }\sqrt{6}$

$=$

$6\sqrt{6}$

c $2\sqrt{14} \times \sqrt{21}$

$14\sqrt{6}$

d $\sqrt{11} \times \sqrt{55}$

$11\sqrt{5}$

7 Simplify the following. Ensure your final answer is fully simplified.
For some questions, it is useful to simplify before multiplying or factorise the surd.

a $2\sqrt{5} \times \sqrt{15}$

$10\sqrt{3}$

b $3\sqrt{7} \times \sqrt{14}$

$21\sqrt{2}$

c $-5\sqrt{10} \times \sqrt{30}$

$-50\sqrt{3}$

d $3\sqrt{6} \times (-\sqrt{18})$

$-18\sqrt{3}$

e $3\sqrt{14} \times 2\sqrt{21}$

$42\sqrt{6}$

f $-2\sqrt{7} \times (-3\sqrt{14})$

$42\sqrt{2}$

g $\sqrt{24} \times \sqrt{20}$

$4\sqrt{30}$

h $\sqrt{54} \times \sqrt{75}$

$45\sqrt{2}$

i $2\sqrt{72} \times 3\sqrt{80}$

$144\sqrt{10}$

j $\sqrt{108} \times (-2\sqrt{125})$

$-60\sqrt{15}$

k $-4\sqrt{27} \times (-\sqrt{28})$

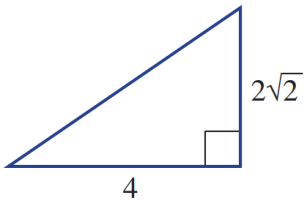
$24\sqrt{21}$

l $\sqrt{98} \times \sqrt{300}$

$70\sqrt{6}$

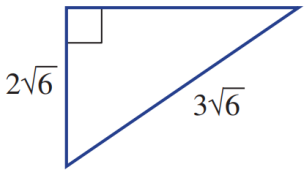
8 Determine the unknown side of the following right-angled triangles.

a



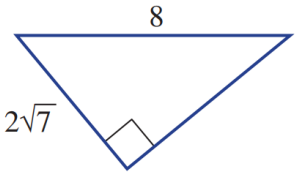
$2\sqrt{6}$

b



$\sqrt{30}$

c



6

9 Find the length of the diagonal of a square, which has an area of 12 cm^2.

A square is a rhombus. The diagonals are equal.

$$A = \frac{xy}{2}$$

$$12 = \frac{x^2}{2}$$

$$x^2 = 24$$

$$x = 2\sqrt{6}$$

Alternate method, each side of the square is $\sqrt{12}$

Using Pythaogras, diagonal is $\sqrt{(\sqrt{12})^2 + (\sqrt{12})^2}$

$$= \sqrt{12 + 12}$$

$$= \sqrt{24}$$

$$= 2\sqrt{6}$$

Dividing Surds

Dividing Surds

$$a\sqrt{x} \div b\sqrt{y} = \frac{a}{b}\sqrt{\frac{x}{y}}$$

1. Divide the coefficients and the radicands separately.
2. Simplify, if possible.

Example Divide surds

$\sqrt{10} \div \sqrt{15}$

1. $\sqrt{\frac{10}{15}}$
2. $= \sqrt{\frac{2}{3}}$

   $= \frac{\sqrt{2}}{\sqrt{3}}$

$18\sqrt{48} \div 6\sqrt{2}$

1. $\frac{18}{6}\sqrt{\frac{48}{2}}$
2. $= 3\sqrt{24}$

   $= 3\sqrt{4}\sqrt{6}$

   $= 6\sqrt{6}$

Guided Practice

Simplify:

a $\sqrt{15} \div \sqrt{3}$ $\sqrt{5}$

b $3\sqrt{10} \div \sqrt{5}$ $3\sqrt{2}$

c $6\sqrt{21} \div 3\sqrt{7}$ $2\sqrt{3}$

d $8\sqrt{11} \div 6\sqrt{25}$ $\frac{4}{15}\sqrt{11}$

e $3\sqrt{80} \div 8\sqrt{5}$ $\frac{3}{2}$

f $\sqrt{2\frac{1}{4}}$ $\frac{3}{2}$

1 Simplify these surds.

a $\sqrt{20} \div \sqrt{2}$

$\sqrt{10}$

b $\sqrt{15} \div \sqrt{3}$

$\sqrt{5}$

c $\sqrt{33} \div (-\sqrt{11})$

$-\sqrt{3}$

d $-\sqrt{30} \div -\sqrt{6}$

$\sqrt{5}$

e $\dfrac{\sqrt{15}}{\sqrt{5}}$

$\sqrt{3}$

f $\dfrac{-\sqrt{40}}{\sqrt{8}}$

$-\sqrt{5}$

2 Simplify these surds.

a $\sqrt{\dfrac{8}{9}}$

$\dfrac{2\sqrt{2}}{3}$

b $\sqrt{\dfrac{12}{49}}$

$\dfrac{2\sqrt{3}}{7}$

c $\sqrt{\dfrac{18}{25}}$

$\dfrac{3\sqrt{2}}{5}$

d $\sqrt{\dfrac{10}{9}}$

$\dfrac{\sqrt{10}}{3}$

e $\sqrt{\dfrac{21}{144}}$

$\dfrac{\sqrt{21}}{12}$

f $\sqrt{\dfrac{26}{32}}$

$\dfrac{\sqrt{13}}{4}$

g $\sqrt{\dfrac{28}{50}}$

$\dfrac{2\sqrt{7}}{5\sqrt{2}}$ or $\dfrac{\sqrt{14}}{5}$

h $\sqrt{\dfrac{15}{27}}$

$\dfrac{\sqrt{5}}{3}$

i $\sqrt{\dfrac{45}{72}}$

$\dfrac{\sqrt{5}}{2\sqrt{2}}$ or $\dfrac{\sqrt{10}}{4}$

3 Simplify the following expressions. Ensure your final answer is fully simplified.

a $\dfrac{\sqrt{45}}{3}$

$\sqrt{5}$

b $\dfrac{\sqrt{24}}{4}$

$\dfrac{\sqrt{6}}{2}$

c $\dfrac{\sqrt{80}}{20}$

$\dfrac{\sqrt{5}}{5}$

d $\dfrac{\sqrt{99}}{18}$

$\dfrac{\sqrt{11}}{6}$

e $\dfrac{3\sqrt{44}}{2}$

$3\sqrt{11}$

f $\dfrac{2\sqrt{98}}{7}$

$2\sqrt{2}$

g $\dfrac{6\sqrt{75}}{20}$

$\dfrac{3\sqrt{3}}{2}$

h $\dfrac{2\sqrt{108}}{18}$

$\dfrac{2\sqrt{3}}{3}$

4 Simplify $\dfrac{\sqrt{12}}{\sqrt{3}}$ by:

a Simplifying each surd then dividing

$\dfrac{2\sqrt{3}}{\sqrt{3}} = 2$

b Dividing the surds then simplifying.

$\sqrt{4} = 2$

5 Simplify the following. Ensure your final answer is fully simplified.
For some questions, it is useful to simplify before dividing.

a $\frac{\sqrt{27}}{\sqrt{3}}$ 3

b $\frac{\sqrt{20}}{\sqrt{5}}$ 2

c $-\frac{\sqrt{162}}{\sqrt{2}}$ -9

d $-\frac{2\sqrt{2}}{5\sqrt{8}}$ $-\frac{1}{5}$

e $\frac{2\sqrt{45}}{15\sqrt{5}}$ $\frac{2}{5}$

f $\frac{5\sqrt{27}}{\sqrt{75}}$ 3

6 Simplify the following.

a $\frac{6\sqrt{14}}{3\sqrt{7}}$ $2\sqrt{2}$

b $\frac{15\sqrt{12}}{5\sqrt{2}}$ $3\sqrt{6}$

c $\frac{4\sqrt{30}}{8\sqrt{6}}$ $\frac{\sqrt{5}}{2}$

d $-\frac{8\sqrt{2}}{2\sqrt{26}}$ $-\frac{4}{\sqrt{13}}$

e $-\frac{3\sqrt{3}}{9\sqrt{21}}$ $-\frac{1}{3\sqrt{7}}$

f $\frac{12\sqrt{70}}{18\sqrt{14}}$ $\frac{2\sqrt{5}}{3}$

7 Simplify fully.

a $\frac{\sqrt{40}}{\sqrt{10}}$ 2

b $\frac{\sqrt{18}}{\sqrt{50}}$ $\frac{3}{5}$

c $\frac{2\sqrt{6}\times\sqrt{5}}{\sqrt{10}}$ $2\sqrt{3}$

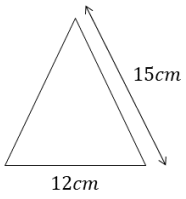
d $\frac{5\sqrt{7}\times\sqrt{3}}{\sqrt{28}}$ $\frac{5\sqrt{3}}{2}$

e $\frac{\sqrt{15}\times\sqrt{20}}{\sqrt{12}}$ 5

f $\frac{6\sqrt{3}\times8\sqrt{2}}{\sqrt{32}\times\sqrt{27}}$ 4

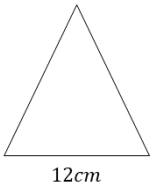
8 Consider the isosceles triangles below.

a Calculate the area of the triangle.



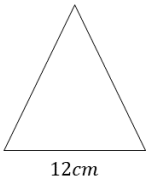
$18\sqrt{21}$ cm²

b The area of the triangle is $18\sqrt{5}$ cm². Calculate the height.



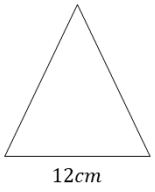
$3\sqrt{5}$ cm

c The area of this triangle is $30\sqrt{5}$ cm². Calculate the perimeter.



$12 + 2\sqrt{161}$ cm

d The perimeter of the triangle is 50 cm. Calculate the area.



$30\sqrt{13}$ cm²

9 Show that $\frac{\sqrt{18y}}{\sqrt{2y}}$ is an integer.

$\frac{\sqrt{18}\sqrt{y}}{\sqrt{2}\sqrt{y}} = \sqrt{9} = 3$

10 Simplify the following.

a $\sqrt{45} + \sqrt{80} - \sqrt{72}$

$7\sqrt{5} - 6\sqrt{2}$

b $3\sqrt{24} + 6\sqrt{150} - 5\sqrt{54}$

$21\sqrt{6}$

c $\frac{4+\sqrt{12}}{2}$

$2 + \sqrt{3}$

d $\frac{2\sqrt{6}\times 3\sqrt{7}}{12\sqrt{21}}$

$\frac{\sqrt{2}}{2}$

11 Simplify the following.

a $(3\sqrt{2})^3$

$= 3^3\sqrt{2}^3$

$= 27 \times 2\sqrt{2}$

$=$

$54\sqrt{2}$

b $(5\sqrt{3})^3$

$375\sqrt{3}$

c $2(3\sqrt{3})^3$

$162\sqrt{3}$

d $(\sqrt{5})^4$

25

e $(-\sqrt{3})^4$

9

f $(2\sqrt{2})^5$

$128\sqrt{2}$

g $-3(2\sqrt{5})^3$

$-120\sqrt{5}$

h $2(-3\sqrt{2})^3$

$-32\sqrt{2}$

i $5(2\sqrt{3})^4$

720

j $\frac{(2\sqrt{7})^3}{4}$

$14\sqrt{7}$

k $\frac{(3\sqrt{2})^3}{4}$

$\frac{27\sqrt{2}}{2}$

l $\frac{(3\sqrt{2})^4}{4}$

81

m $\frac{(5\sqrt{2})^2}{4} \times \frac{(2\sqrt{3})^3}{3}$

$100\sqrt{3}$

n $\frac{(2\sqrt{3})^2}{9} \times \frac{(-3\sqrt{2})^4}{3}$

144

o $\frac{(2\sqrt{5})^3}{5} \times \frac{(-2\sqrt{3})^5}{24}$

$-96\sqrt{15}$

p $\frac{(3\sqrt{3})^3}{2} \div \frac{(5\sqrt{2})^2}{4}$

$\frac{81\sqrt{3}}{25}$

q $\frac{(2\sqrt{5})^4}{50} \div \frac{(2\sqrt{3})^3}{5}$

$\frac{5}{3\sqrt{3}}$

r $\frac{(2\sqrt{2})^3}{9} \div \frac{(2\sqrt{8})^2}{(\sqrt{27})^3}$

$\frac{9\sqrt{6}}{2}$

Distributive Law with Surds

Expanding Brackets with Surds

$$a(b+c) = ab + ac$$

1. Expand brackets.
2. Simplify.

Example Expand brackets with surds

$\sqrt{3}(3\sqrt{5}+\sqrt{6})$	$3\sqrt{6}(2\sqrt{10}-4\sqrt{6})$
1. $3\sqrt{15}+\sqrt{18}$	1. $6\sqrt{60}-12\sqrt{36}$
2. $3\sqrt{15}+3\sqrt{2}$	2. $12\sqrt{15}-72$

Guided Practice

Simplify:

a $\sqrt{2}(\sqrt{10}-\sqrt{2})$ $2\sqrt{5}-2$	**b** $\sqrt{6}(\sqrt{8}-2)$ $4\sqrt{3}-2\sqrt{6}$	**c** $\sqrt{3}(\sqrt{2}+\sqrt{5})$ $\sqrt{6}+\sqrt{15}$
d $2\sqrt{3}(\sqrt{5}+\sqrt{7})$ $2\sqrt{15}+2\sqrt{21}$	**e** $3\sqrt{2}(2\sqrt{13}-\sqrt{11})$ $6\sqrt{26}-3\sqrt{22}$	**f** $4\sqrt{5}(\sqrt{5}-\sqrt{10})$ $20-20\sqrt{2}$

1 Use the distributive law to expand the following and simplify the surds where necessary.

a $\sqrt{2}(\sqrt{10}-\sqrt{2})$

$2\sqrt{5}-2$

b $\sqrt{6}(3+\sqrt{3})$

$3\sqrt{6}+3\sqrt{2}$

c $\sqrt{5}(\sqrt{15}+4)$

$5\sqrt{3}+4\sqrt{5}$

d $\sqrt{6}(\sqrt{8}-2)$

$4\sqrt{3}-2\sqrt{6}$

e $3\sqrt{3}(9-\sqrt{21})$

$27\sqrt{3}-9\sqrt{7}$

f $3\sqrt{7}(\sqrt{14}-2\sqrt{7})$

$21\sqrt{2}-42$

g $-2\sqrt{6}(3\sqrt{2}-2\sqrt{3})$

$-12\sqrt{3}+12\sqrt{2}$

h $3\sqrt{7}(2\sqrt{7}+3\sqrt{14})$

$42+63\sqrt{2}$

i $-2\sqrt{8}(2\sqrt{2}-3\sqrt{20})$

$-16+24\sqrt{10}$

j $2\sqrt{3}(7\sqrt{6}+5\sqrt{3})$

$42\sqrt{2}+30$

2 Expand and simplify the following.

a $\sqrt{2}(3-\sqrt{3})-\sqrt{8}$

$\sqrt{2}-\sqrt{6}$

b $\sqrt{8}(\sqrt{6}+\sqrt{2})-\sqrt{3}$

$3\sqrt{3}+4$

c $\sqrt{5}(\sqrt{2}+1)-\sqrt{40}$

$\sqrt{5}-\sqrt{10}$

d $\sqrt{44}-2(\sqrt{11}-1)$

2

e $\sqrt{24}-2\sqrt{2}(\sqrt{3}-4)$

$8\sqrt{2}$

f $2\sqrt{3}(\sqrt{6}-\sqrt{3})-\sqrt{50}$

$\sqrt{2}-6$

Expanding Binomials with Surds

Expanding Brackets with Surds

$$(a+b)(c+d) = ac + ad + bc + bd$$

1. Expand brackets.
2. Simplify.

Example Expand brackets with surds

$(\sqrt{15}+2)(\sqrt{3}-3)$

$= \sqrt{45} - 3\sqrt{15} + 2\sqrt{3} - 6$

$= 3\sqrt{5} - 3\sqrt{15} + 2\sqrt{3} - 6$

$(\sqrt{15}-\sqrt{6})^2$

$= 15 - 2\sqrt{90} + 6$

$= 21 - 2\sqrt{90}$

$= 21 - 6\sqrt{10}$

Guided Practice

Simplify:

a $(\sqrt{3}+1)(\sqrt{2}-1)$

$\sqrt{6} - \sqrt{3} + \sqrt{2} - 1$

b $(\sqrt{5}+1)(\sqrt{5}-1)$

4

c $(\sqrt{7}-\sqrt{5})^2$

$12 - 2\sqrt{35}$

d $(2\sqrt{6}-1)(\sqrt{6}-2)$

$14 - 5\sqrt{6}$

e $(5\sqrt{2}-1)(3\sqrt{2}+3)$

$27 + 12\sqrt{2}$

f $(\sqrt{13}+\sqrt{9})^2$

$22 + 6\sqrt{13}$

1 Expand these binomial expressions.

a $(\sqrt{3}+1)(\sqrt{2}-1)$

$\sqrt{6}+\sqrt{2}-\sqrt{3}-1$

b $(\sqrt{5}-2)(\sqrt{7}+3)$

$\sqrt{35}+3\sqrt{5}-2\sqrt{7}-6$

c $(\sqrt{5}+\sqrt{2})(\sqrt{3}+\sqrt{2})$

$\sqrt{15}+\sqrt{10}+\sqrt{6}+2$

d $(\sqrt{6}-1)(\sqrt{6}-2)$

$8-3\sqrt{6}$

e $(\sqrt{7}-2)(2\sqrt{7}+5)$

$4+\sqrt{7}$

f $(3\sqrt{2}-1)(\sqrt{6}-\sqrt{3})$

$7\sqrt{3}-4\sqrt{6}$

2 Expand these expressions.

a $(3-2\sqrt{7})(\sqrt{21}-4\sqrt{3})$

$11\sqrt{21}-26\sqrt{3}$

b $(2\sqrt{6}+5)(\sqrt{30}-2\sqrt{5})$

$2\sqrt{5}+\sqrt{30}$

c $(3\sqrt{5}+1)(\sqrt{7}+2\sqrt{35})$

$30+\sqrt{7}+5\sqrt{35}$

d $(4\sqrt{2}+\sqrt{7})(3\sqrt{14}-5)$

$19\sqrt{7}+\sqrt{2}$

e $(3\sqrt{3}+4)(\sqrt{6}-2\sqrt{2})$

$\sqrt{2}-2\sqrt{6}$

f $(5-3\sqrt{2})(2\sqrt{10}+3\sqrt{5})$

$\sqrt{10}+3\sqrt{5}$

3 Expand these expressions.

a $(\sqrt{5}+2)(\sqrt{5}+1)$

$7+3\sqrt{5}$

b $(2+\sqrt{5})(1+\sqrt{5})$

$7+3\sqrt{5}$

c $(2-\sqrt{5})(1+\sqrt{5})$

$-3+\sqrt{5}$

d $(\sqrt{5}-2)(\sqrt{5}-1)$

$7-3\sqrt{5}$

e $(2+2\sqrt{5})(1+\sqrt{5})$

$12+4\sqrt{5}$

f $(\sqrt{5}+2)(\sqrt{5}-1)$

$3+\sqrt{5}$

g $(\sqrt{7}+2)(\sqrt{7}-1)$

$5+\sqrt{7}$

h $(2\sqrt{7}+2)(3\sqrt{7}-1)$

$40+4\sqrt{7}$

i $(2\sqrt{7}+2)(3\sqrt{7}-6)$

$30-6\sqrt{7}$

j $(\sqrt{7}+\sqrt{6})(\sqrt{7}-\sqrt{6})$

1

k $(\sqrt{7}-\sqrt{6})(\sqrt{7}-\sqrt{6})$

$13-2\sqrt{42}$

l $(\sqrt{7}+\sqrt{6})^2$

$13+2\sqrt{2}$

4 Expand these perfect squares.

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

a	$(\sqrt{3}+1)^2$	**b**	$(2-\sqrt{6})^2$	**c**	$(4+\sqrt{7})^2$
	$4+2\sqrt{3}$		$10-4\sqrt{6}$		$23+8\sqrt{7}$
d	$(2\sqrt{3}+4)^2$	**e**	$(2\sqrt{7}+\sqrt{5})^2$	**f**	$(3\sqrt{2}-2\sqrt{3})^2$
	$28+16\sqrt{3}$		$33+4\sqrt{35}$		$30-12\sqrt{6}$
g	$(5\sqrt{3}-2\sqrt{8})^2$	**h**	$(3\sqrt{5}+4\sqrt{7})^2$	**i**	$(5\sqrt{6}+3\sqrt{5})^2$
	$107-40\sqrt{6}$		$157+24\sqrt{35}$		$195+30\sqrt{30}$

5 Expand these using the difference of two squares.

$$(a+b)(a-b) = a^2 - b^2$$

a	$(3-\sqrt{2})(3+\sqrt{2})$	**b**	$(5-\sqrt{6})(5+\sqrt{6})$	**c**	$(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})$
	7		19		1
d	$(3\sqrt{2}-\sqrt{11})(3\sqrt{2}+\sqrt{11})$	**e**	$(2\sqrt{6}+3)(2\sqrt{6}-3)$	**f**	$(7-2\sqrt{5})(7+2\sqrt{5})$
	7		15		29
g	$(5\sqrt{8}-10\sqrt{2})(5\sqrt{8}+10\sqrt{2})$	**h**	$(3\sqrt{7}-4\sqrt{6})(3\sqrt{7}+4\sqrt{6})$		
	0		-33		
i	$(2\sqrt{10}+4\sqrt{5})(2\sqrt{10}-4\sqrt{5})$	**j**	$(4\sqrt{10}-5\sqrt{6})(4\sqrt{10}+5\sqrt{6})$		
	-40		10		

6 What do you notice about the result of a difference of two squares above?

The answer has no surds.

7 Simplify the following expressions.

a 5×5

25

b $5 \times \sqrt{5}$

$5\sqrt{5}$

c $\sqrt{5} \times \sqrt{5}$

5

d $5 \times 5\sqrt{5}$

$25\sqrt{5}$

e $\sqrt{5} \times 5\sqrt{5}$

25

f $5\sqrt{5} \times 5\sqrt{5}$

125

g $5(5 + \sqrt{5})$

$25 + 5\sqrt{5}$

h $\sqrt{5}(5 + \sqrt{5})$

$5 + 5\sqrt{5}$

i $5(5 + 5\sqrt{5})$

$25 + 25\sqrt{5}$

j $5\sqrt{5}(5 + \sqrt{5})$

$25 + 25\sqrt{5}$

k $5\sqrt{5}(\sqrt{5} + 5\sqrt{5})$

150

l $(5 + \sqrt{5})(5 + \sqrt{5})$

$30 + 10\sqrt{5}$

m $(5 + \sqrt{5})(5 - \sqrt{5})$

20

n $(5 + \sqrt{5})(5 + 5\sqrt{5})$

$50 + 30\sqrt{5}$

o $(5 + 5\sqrt{5})(5 + 5\sqrt{5})$

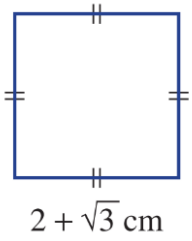
$150 + 50\sqrt{5}$

p $(\sqrt{5} + 5\sqrt{5})(\sqrt{5} + \sqrt{5})$

60

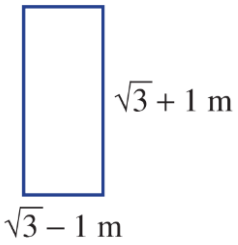
8 Find the area of these shapes in exact form.

a



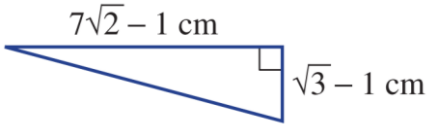
$7 + 4\sqrt{3}$

b



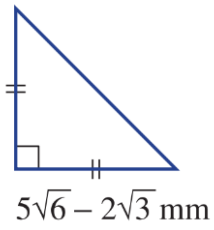
2

c



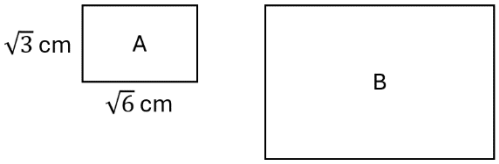
$\frac{1}{2}(7\sqrt{6} - 7\sqrt{2} - \sqrt{3} + 1)$

d



$81 - 30\sqrt{2}$

9 These rectangles are similar. The scale factor is 4.
Find the area of B in its simplest form.



$48\sqrt{2}$ cm^2

10 Simplify the following.

a $\left(1+\frac{1}{\sqrt{2}}\right)^2$

$\frac{3}{2}+\frac{2}{\sqrt{2}}$

b $\left(\sqrt{3}-\frac{1}{\sqrt{3}}\right)^2$

$\frac{4}{3}$

11 Show that $x=-3+2\sqrt{2}$ satisfies the equation $x^2+6x+1=0$

$LHS=\left(-3+2\sqrt{2}\right)^2+6\left(-3+2\sqrt{2}\right)+1$

$=9-12\sqrt{2}+8-18+12\sqrt{2}+1$

$=0=RHS$

12 Find the value of a and b.

a $\left(\sqrt{3}+1\right)^2=a+b\sqrt{3}$

$a=4, b=2$

b $\left(4-\sqrt{5}\right)^2=a+b\sqrt{5}$

$a=21, b=-8$

c $\left(2\sqrt{3}+\sqrt{2}\right)^2=a+b\sqrt{6}$

$a=14, b=4$

d $\left(2\sqrt{2}+1\right)^2=a+4\sqrt{b}$

$a=9, b=2$

e $\left(2\sqrt{3}-3\sqrt{6}\right)^2=a+b\sqrt{2}$

$a=66, b=-12$

f $\left(1-3\sqrt{2}\right)^2-\left(2+3\sqrt{2}\right)^2=a+b\sqrt{2}$

$a=-3, b=-18$

Rationalising the Denominator

Rationalising the Denominator

- Rationalising the denominator means to write a fraction with an integer denominator.
- Multiply the numerator and denominator by the surd in the denominator.

$\frac{a}{\sqrt{b}} \times \frac{\sqrt{b}}{\sqrt{b}} = \frac{a\sqrt{b}}{b}$	$\frac{a}{b\sqrt{c}} \times \frac{\sqrt{c}}{\sqrt{c}} = \frac{a\sqrt{c}}{bc}$	$\frac{a\sqrt{b}}{c\sqrt{d}} \times \frac{\sqrt{d}}{\sqrt{d}} = \frac{a\sqrt{bd}}{cd}$

Example Success Criteria

$\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$ $= \frac{\sqrt{2}}{2}$	$\frac{2}{5\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{2\sqrt{3}}{15}$

Guided Practice

Rationalise these expressions.

a $\frac{1}{\sqrt{3}}$ $\frac{\sqrt{3}}{3}$	**b** $\frac{\sqrt{2}}{\sqrt{3}}$ $\frac{\sqrt{6}}{3}$	**c** $\frac{1+\sqrt{2}}{\sqrt{3}}$ $\frac{\sqrt{3}+\sqrt{6}}{3}$
d $\frac{\sqrt{6}-5}{\sqrt{2}}$ $\frac{2\sqrt{3}-5\sqrt{2}}{2}$	**e** $\frac{\sqrt{3}}{2\sqrt{2}}$ $\frac{\sqrt{6}}{4}$	**f** $\frac{8+3\sqrt{2}}{4\sqrt{5}}$ $\frac{8\sqrt{5}+3\sqrt{10}}{20}$

Key Ideas

1. Rationalising the denominator means to write the denominator as an ……
2. For $\frac{a\sqrt{b}}{c\sqrt{d}}$, we rationalise the denominator by multiplying …… and …… by ……

1 Write the missing number.

a	$\sqrt{3} \times \ldots\ldots = 3$	**b**	$\ldots\ldots \times \sqrt{5} = 5$	**c**	$2\sqrt{5} \times \ldots\ldots = 10$
	$\sqrt{3}$		$\sqrt{5}$		$\sqrt{5}$
d	$4\sqrt{3} \times \ldots\ldots = 12$	**e**	$\ldots\ldots \times \sqrt{7} = 21$	**f**	$3\sqrt{6} \times \ldots\ldots = 18$
	$\sqrt{3}$		$3\sqrt{7}$		$\sqrt{6}$
g	$\frac{1}{\sqrt{3}} \times \text{—} = \frac{\sqrt{3}}{3}$	**h**	$\frac{1}{\sqrt{7}} \times \text{—} = \frac{\sqrt{7}}{7}$	**i**	$\frac{2}{\sqrt{13}} \times \text{—} = \frac{2\sqrt{13}}{13}$
	$\frac{\sqrt{3}}{\sqrt{3}}$		$\frac{\sqrt{7}}{\sqrt{7}}$		$\frac{\sqrt{13}}{\sqrt{13}}$

2 Rationalise these denominators.

a	$\frac{1}{\sqrt{2}}$	**b**	$\frac{1}{\sqrt{7}}$	**c**	$\frac{3}{\sqrt{11}}$
	$\frac{\sqrt{2}}{2}$		$\frac{\sqrt{7}}{7}$		$\frac{3\sqrt{11}}{11}$
d	$\frac{4}{\sqrt{5}}$	**e**	$\frac{\sqrt{5}}{\sqrt{3}}$	**f**	$\frac{\sqrt{2}}{\sqrt{7}}$
	$\frac{4\sqrt{5}}{\sqrt{5}}$		$\frac{\sqrt{15}}{3}$		$\frac{\sqrt{14}}{7}$

3 Rewrite the following in the form $\frac{\sqrt{a}}{\sqrt{b}}$ and then rationalise the denominators.

a	$\sqrt{\frac{2}{3}}$	**b**	$\sqrt{\frac{5}{7}}$	**c**	$\sqrt{\frac{2}{5}}$
	$\frac{\sqrt{6}}{3}$		$\frac{\sqrt{35}}{7}$		$\frac{\sqrt{10}}{5}$
d	$\sqrt{\frac{7}{3}}$	**e**	$\sqrt{\frac{6}{7}}$	**f**	$\sqrt{\frac{17}{2}}$
	$\frac{\sqrt{21}}{3}$		$\frac{\sqrt{42}}{7}$		$\frac{\sqrt{34}}{2}$

4 Rationalise these denominators.

a	$\frac{4\sqrt{2}}{\sqrt{7}}$	**b**	$\frac{5\sqrt{2}}{\sqrt{3}}$	**c**	$\frac{3\sqrt{5}}{\sqrt{2}}$
	$\frac{4\sqrt{14}}{7}$		$\frac{5\sqrt{6}}{3}$		$\frac{3\sqrt{10}}{2}$
d	$\frac{3\sqrt{6}}{\sqrt{7}}$	**e**	$\frac{7\sqrt{3}}{\sqrt{10}}$	**f**	$\frac{2\sqrt{7}}{\sqrt{15}}$
	$\frac{3\sqrt{42}}{7}$		$\frac{7\sqrt{30}}{10}$		$\frac{2\sqrt{105}}{15}$

5 By rationalising the denominator, show that $\frac{n}{\sqrt{n}} = \sqrt{n}$.

$\frac{n\sqrt{n}}{n} = \sqrt{n}$

Therefore, simplify:

a	$\frac{2}{\sqrt{2}}$	**b**	$\frac{5}{\sqrt{5}}$	**c**	$\frac{93}{\sqrt{93}}$	**d**	$\frac{ab}{\sqrt{ab}}$
	$\sqrt{2}$		$\sqrt{5}$		$\sqrt{93}$		$\sqrt{ab}$

6 Rationalise these denominators.

a $\frac{5}{\sqrt{2}}$

$\frac{5\sqrt{2}}{2}$

b $\frac{6}{\sqrt{2}}$

$3\sqrt{2}$

c $\frac{6}{\sqrt{3}}$

$2\sqrt{3}$

d $\frac{4}{\sqrt{3}}$

$\frac{4\sqrt{3}}{3}$

e $\frac{4}{\sqrt{6}}$

$\frac{2\sqrt{6}}{3}$

f $\frac{5}{\sqrt{6}}$

$\frac{5\sqrt{6}}{6}$

g $\frac{5}{2\sqrt{6}}$

$\frac{5\sqrt{6}}{12}$

h $\frac{5}{3\sqrt{6}}$

$\frac{5\sqrt{6}}{18}$

i $\frac{5}{10\sqrt{6}}$

$\frac{\sqrt{6}}{12}$

j $\frac{5}{10\sqrt{5}}$

$\frac{\sqrt{5}}{10}$

k $\frac{5}{\sqrt{5}}$

$\sqrt{5}$

l $\frac{8}{\sqrt{8}}$

$\sqrt{8}$

m $\frac{10}{\sqrt{8}}$

$\frac{5\sqrt{8}}{4}$

n $\frac{10}{\sqrt{12}}$

$\frac{5\sqrt{12}}{6}$

o $\frac{10}{2\sqrt{3}}$

$\frac{5\sqrt{3}}{3}$

7 Rationalise these denominators.

a $\frac{4\sqrt{7}}{5\sqrt{3}}$

$\frac{4\sqrt{21}}{15}$

b $\frac{2\sqrt{3}}{3\sqrt{2}}$

$\frac{\sqrt{6}}{3}$

c $\frac{5\sqrt{7}}{3\sqrt{5}}$

$\frac{\sqrt{35}}{3}$

d $\frac{5\sqrt{12}}{3\sqrt{27}}$

$\frac{10}{9}$

e $\frac{9\sqrt{6}}{2\sqrt{3}}$

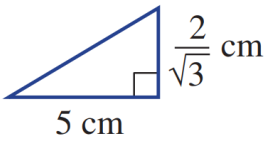
$\frac{9\sqrt{2}}{2}$

f $\frac{7\sqrt{90}}{2\sqrt{70}}$

$\frac{3\sqrt{7}}{2}$

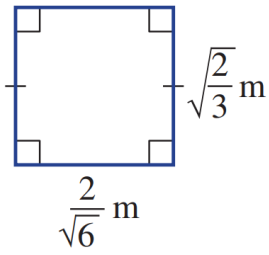
8 Determine the exact value of the area of the following shapes.
Express your answers using a rational denominator.

a



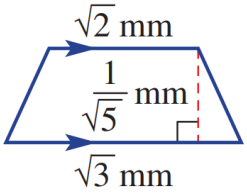
$\frac{5\sqrt{3}}{3}$

b



$\frac{2}{3}$

c



$\frac{\sqrt{10}+\sqrt{15}}{10}$

9 Rationalise these denominators.

a $\frac{1+\sqrt{2}}{\sqrt{3}}$

$\frac{\sqrt{3}+\sqrt{6}}{3}$

b $\frac{2-\sqrt{3}}{\sqrt{5}}$

$\frac{2\sqrt{5}-\sqrt{15}}{5}$

c $\frac{\sqrt{5}+\sqrt{2}}{\sqrt{7}}$

$\frac{\sqrt{35}+\sqrt{14}}{7}$

d $\frac{\sqrt{2}+\sqrt{7}}{\sqrt{6}}$

$\frac{2\sqrt{3}+\sqrt{42}}{6}$

e $\frac{6-\sqrt{10}}{\sqrt{5}}$

$\frac{6\sqrt{5}-5\sqrt{2}}{5}$

f $\frac{3\sqrt{5}+5\sqrt{2}}{\sqrt{10}}$

$\frac{3\sqrt{2}+2\sqrt{5}}{2}$

13 Simplify the following.

$$xy\left(\frac{\sqrt{x}}{\sqrt{y}}-\frac{\sqrt{y}}{\sqrt{x}}\right)^2$$

$(x-y)^2$

Rationalising Binomial Denominators

ⓘ The Conjugate

The **conjugate** of $a + b\sqrt{c}$ is $a - b\sqrt{c}$.

🔍 Identify the conjugate

Expression	Conjugate
$7 + \sqrt{3}$	
$8 + \sqrt{5}$	
$\sqrt{5} + 8$	
$\sqrt{5} - 8$	
$4 - \sqrt{7}$	
$5 + 3\sqrt{2}$	
$6 - 2\sqrt{3}$	
$4\sqrt{6} - 3$	

ⓘ The Conjugate Product

- The product of an expression and its conjugate is a difference of two squares.
- A conjugate product has no surds.

$(a + \sqrt{c})(a - \sqrt{c})$ $= a^2 - c$	$(a + b\sqrt{c})(a - b\sqrt{c})$ $= a^2 - b^2c$

🔍 Identify the conjugate product

a $(2 + \sqrt{3})(2 - \sqrt{3})$

$= 2^2 - (\sqrt{3})^2$

$= 4 - 3$

$= 1$

b $(5 - \sqrt{6})(5 + \sqrt{6})$

19

c $(7 - \sqrt{3})(7 + \sqrt{3})$

46

d $(5 + 3\sqrt{6})(5 - 3\sqrt{6})$

−29

e $(7 + 2\sqrt{5})(7 - 2\sqrt{5})$

29

f $(12 - 2\sqrt{3})(12 + \sqrt{3})$

132

Rationalising Binomial Denominator

For binomial denominators, rationalise by **multiplying by the conjugate**.

$$\frac{a}{\sqrt{b}+\sqrt{c}} \times \frac{\sqrt{b}-\sqrt{c}}{\sqrt{b}-\sqrt{c}} = \frac{a\sqrt{b}-a\sqrt{c}}{b^2-c^2}$$

$$\frac{a}{\sqrt{b}-\sqrt{c}} \times \frac{\sqrt{b}+\sqrt{c}}{\sqrt{b}+\sqrt{c}} = \frac{a\sqrt{b}+a\sqrt{c}}{b^2-c^2}$$

Example Rationalise a binomial denominator

$$\frac{3}{5+\sqrt{3}} \times \frac{5-\sqrt{3}}{5-\sqrt{3}}$$

$$= \frac{15-3\sqrt{3}}{25-3}$$

$$= \frac{15-3\sqrt{3}}{22}$$

$$\frac{\sqrt{5}}{2\sqrt{2}-3} \times \frac{2\sqrt{2}+3}{2\sqrt{2}+3}$$

$$= \frac{2\sqrt{10}+3\sqrt{5}}{4\times 2-9}$$

$$= -\left(2\sqrt{10}+3\sqrt{5}\right)$$

Guided Practice

Rationalise these expressions.

a $\frac{1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$

$\frac{\sqrt{3}+1}{2}$

b $\frac{1}{\sqrt{5}-\sqrt{2}}$

$\frac{\sqrt{5}+\sqrt{2}}{3}$

c $\frac{4}{\sqrt{3}+\sqrt{2}}$

$4\sqrt{3}-4\sqrt{2}$

d $\frac{\sqrt{3}-4}{\sqrt{3}+4}$

$\frac{-19+8\sqrt{3}}{13}$

e $\frac{2\sqrt{7}}{2\sqrt{7}-5}$

$\frac{28+10\sqrt{7}}{3}$

f $\frac{2\sqrt{3}}{\sqrt{5}+2\sqrt{6}}$

$\frac{-2\sqrt{15}+12\sqrt{2}}{19}$

Key Ideas

1. To rationalise expressions with binomial denominators, we multiply by the ……
2. The conjugate of $a+b$ is ……, and the conjugate of $a-b$ is ……

1 Expand these products of conjugate pairs.

a $(4-\sqrt{3})(4+\sqrt{3})$

13

b $(5+\sqrt{7})(5-\sqrt{7})$

18

c $(\sqrt{6}-2\sqrt{10})(\sqrt{6}+2\sqrt{10})$

-34

d $(a+\sqrt{b})(a-\sqrt{b})$

a^2-b

Do you get a rational number as a result in each example?

yes

2 Why is $\frac{2}{4+\sqrt{3}}$ the same value as $\frac{2}{4+\sqrt{3}} \times \frac{4-\sqrt{3}}{4-\sqrt{3}}$?

You multiply the numerator and denominator by the same expression, so the resulting expression is equivalent.

Rationalise these denominators.

a $\frac{2}{4+\sqrt{3}}$

$\frac{8-2\sqrt{3}}{13}$

b $\frac{1}{5-\sqrt{7}}$

$\frac{5+\sqrt{7}}{9}$

c $\frac{\sqrt{5}}{8-\sqrt{5}}$

$\frac{8\sqrt{5}+5}{59}$

d $\frac{5}{\sqrt{6}-1}$

$\sqrt{6}+1$

e $\frac{6}{\sqrt{2}+\sqrt{11}}$

$\frac{2\sqrt{11}-2\sqrt{2}}{3}$

f $\frac{2+\sqrt{3}}{2-\sqrt{3}}$

$7+4\sqrt{3}$

3 The expression $\frac{\sqrt{6}+1}{\sqrt{3}+\sqrt{2}}$ can be written as $a\sqrt{3}+b\sqrt{2}$. Find a and b.

$a=-1, b=2$

10 Rationalise these denominators.

a $\frac{2}{\sqrt{2}+3}$

$6-2\sqrt{2}$

b $\frac{2}{3+\sqrt{2}}$

$\frac{6-2\sqrt{2}}{7}$

c $\frac{2}{3-\sqrt{2}}$

$\frac{6+2\sqrt{2}}{7}$

d $\frac{7}{3-\sqrt{2}}$

$3+\sqrt{2}$

e $\frac{7}{\sqrt{2}-3}$

$-\sqrt{2}-3$

f $\frac{7}{\sqrt{7}-3}$

$\frac{-7\sqrt{7}-21}{2}$

g $\frac{\sqrt{7}}{\sqrt{7}-3}$

$\frac{-7-3\sqrt{7}}{2}$

h $\frac{\sqrt{5}}{\sqrt{5}-3}$

$\frac{-5-3\sqrt{5}}{4}$

i $\frac{\sqrt{5}}{5-\sqrt{3}}$

$\frac{5\sqrt{5}+\sqrt{15}}{22}$

j $\frac{2\sqrt{5}}{5-\sqrt{3}}$

$\frac{5\sqrt{5}+\sqrt{15}}{11}$

k $\frac{4\sqrt{5}}{5-\sqrt{3}}$

$\frac{10\sqrt{5}+2\sqrt{15}}{11}$

l $\frac{4\sqrt{5}}{5-2\sqrt{3}}$

$\frac{20\sqrt{5}+8\sqrt{15}}{13}$

m $\frac{4\sqrt{5}}{5-3\sqrt{3}}$

$-10\sqrt{5}-6\sqrt{15}$

n $\frac{4\sqrt{3}}{5-3\sqrt{3}}$

$-10\sqrt{3}-18$

o $\frac{4\sqrt{3}-\sqrt{2}}{5-3\sqrt{3}}$

$\frac{-20\sqrt{3}-36+5\sqrt{2}+3\sqrt{6}}{2}$

4 Simplify by forming the lowest common denominator:

a $\frac{1}{\sqrt{3}+1}+\frac{1}{\sqrt{3}-1}$

$\sqrt{3}$

b $\frac{3}{2\sqrt{5}-\sqrt{7}}-\frac{3}{2\sqrt{5}+\sqrt{7}}$

$\frac{6\sqrt{7}}{13}$

5 Show that the following expression is rational:

$$\frac{4}{2+\sqrt{5}}-\frac{1}{9-4\sqrt{5}}$$

$$=\frac{4}{2+\sqrt{5}}\times\frac{(2-\sqrt{5})}{(2-\sqrt{5})}-\frac{1}{9-4\sqrt{5}}\times\frac{(9+4\sqrt{5})}{(9+4\sqrt{5})}$$

$$=\frac{8-4\sqrt{5}}{-1}-\frac{9+4\sqrt{5}}{81-80}$$

$$=-17$$

6 Simplify the following by rationalising the denominator.

a $\frac{5+\sqrt{2}}{5-\sqrt{2}}$

$\frac{27+10\sqrt{2}}{23}$

b $\frac{1}{\sqrt{3}+1}+\frac{\sqrt{3}-1}{4}$

$\frac{3\sqrt{3}-3}{4}$

c $\frac{42}{\sqrt{63}}-\frac{6}{\sqrt{7}+2}$

4

7 Simplify the following by rationalising the denominator.

a $\frac{a\sqrt{b}}{c\sqrt{d}}$

$\frac{a\sqrt{bd}}{cd}$

b $\frac{1}{a+\sqrt{b}}$

$\frac{a-\sqrt{b}}{a^2-b}$

c $\frac{1}{\sqrt{a}}-\frac{1}{\sqrt{b}}$

$\frac{\sqrt{a}}{a}-\frac{\sqrt{b}}{b}=\frac{b\sqrt{a}-a\sqrt{b}}{ab}$

d $\frac{1}{a\sqrt{b}-c}$

$\frac{a\sqrt{b}+c}{a^2b-c^2}$

e $\frac{1}{a\sqrt{b}+c\sqrt{d}}$

$\frac{a\sqrt{b}-c\sqrt{d}}{a^2b-c^2d}$

8 Show these expressions are rational by rationalising the denominator.

a $\frac{1}{3+\sqrt{6}}+\frac{2}{\sqrt{6}}$

1

b $\frac{4}{2+\sqrt{2}}+\frac{1}{3-2\sqrt{2}}$

7

c $\frac{8}{3-\sqrt{7}}-\frac{6}{2\sqrt{7}-5}$

2

9

a Rationalise the denominator of

$$\frac{1}{\sqrt{n}+\sqrt{n+1}}$$

$$\sqrt{n+1}-\sqrt{n}$$

b Hence simplify

$$\frac{1}{1+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}$$

$$=(\sqrt{2}-1)+(\sqrt{3}-\sqrt{2})+(\sqrt{4}-\sqrt{3})+\cdots(\sqrt{100}-\sqrt{99})$$

$$=-1+\sqrt{100}=9$$

10

a Expand $\left(x+\frac{1}{x}\right)^2$

$$x^2+2+\frac{1}{x^2}$$

b Suppose that $x=\sqrt{7}+\sqrt{6}$

i Show that $x+\frac{1}{x}=2\sqrt{7}$

$$\sqrt{7}+\sqrt{6}+\frac{1}{\sqrt{7}+\sqrt{6}}$$

$$=\sqrt{7}+\sqrt{6}+\frac{\sqrt{7}-\sqrt{6}}{1}$$

$$=2\sqrt{7}$$

ii Use the result in part **a** to find the value of $x^2+\frac{1}{x^2}$

$$\left(x+\frac{1}{x}\right)^2=x^2+2+\frac{1}{x^2}$$

$$\therefore x^2+\frac{1}{x^2}=\left(x+\frac{1}{x}\right)^2-2$$

$$=\left(2\sqrt{7}\right)^2-2$$

$$=26$$

11 Find the value of $\sqrt{7-4\sqrt{3}}\times\sqrt{7+4\sqrt{3}}$

$$=\sqrt{(7-4\sqrt{3})(7+4\sqrt{3})}$$

$$=\sqrt{49-48}$$

$$=1$$

12

a Write down the expansion of $(a-b)^2$

$a^2-2ab+b^2$

b Use the expansion in part **a** to square $\sqrt{6+\sqrt{11}}-\sqrt{6-\sqrt{11}}$.

Let $a=\sqrt{6+\sqrt{11}}, b=\sqrt{6-\sqrt{11}}$

$$(a-b)^2=(6+\sqrt{11})-2\sqrt{6+\sqrt{11}}\sqrt{6-\sqrt{11}}+(6-\sqrt{11})$$

$$=12-2\sqrt{(6+\sqrt{11})(6-\sqrt{11})}$$

$$=12-2\sqrt{36-11}$$

$$=12-2\sqrt{25}$$

$$=2$$

c Hence evaluate $\sqrt{6+\sqrt{11}}-\sqrt{6-\sqrt{11}}$

$\sqrt{2}$

13 If $x=\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1}}}}$..., find the value of x.

Notice that since this repeats infinitely, we can write as $x=\sqrt{1+x}$

Square both sides

$$x^2=1+x$$

Solve using quadratic formula

$$x=\frac{1\pm\sqrt{5}}{2}$$

x must be positive as it is a root, so

$$x=\frac{1+\sqrt{5}}{2}$$

(this number $\phi\approx1.618$ is the golden ratio)

14 If $\sqrt{x+\sqrt{x^2-4}}+\sqrt{x-\sqrt{x^2-4}}=4$, find the value of x.

Let $a=\sqrt{x+\sqrt{x^2-4}}$, $b=\sqrt{x-\sqrt{x^2-4}}$

It is given that $a+b=4$

$$ab=\sqrt{\left(x+\sqrt{x^2-4}\right)\left(x-\sqrt{x^2-4}\right)}$$

$$=\sqrt{x^2-(x^2-4)}$$

$$=\sqrt{4}=2$$

$$a+b=4$$

$$ab=2$$

Solving simultaneously,

$$a(4-a)=2$$

$$a^2-4a+2=0$$

$a=2-\sqrt{2}$ or $2+\sqrt{2}$

You can similarly find values of b:

$b=2+\sqrt{2}$ or $2-\sqrt{2}$

Using values of a:

$$\sqrt{x+\sqrt{x^2-4}}=2+\sqrt{2}$$

$$x+\sqrt{x^2-4}=6+4\sqrt{2}$$

$$x+\sqrt{x^2-4}=6+\sqrt{32}$$

By equating coefficients, $x=6$

$$\sqrt{x+\sqrt{x^2-4}}=2-\sqrt{2}$$

$$x+\sqrt{x^2-4}=6-4\sqrt{2}$$

$$x+\sqrt{x^2-4}=6-\sqrt{32}$$

No solution.

You can repeat for values of b, the only solutions is again $x=6$

Therefore, $x=6$